



OIL INDUSTRY SAFETY DIRECTORATE

**तटीय और अपतटीय ड्रिलिंग और वर्कओवर रिगों में विद्युत सुरक्षा
ओआईएसडी- मानक- 216**

**ELECTRICAL SAFETY IN ONSHORE AND
OFFSHORE DRILLING AND WORKOVER RIGS**

OISD-STANDARD- 216

Inception	September 2007
Amended Edition	October 2010
Amended Edition	January 2015
1 st Revision	*****2026

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OIL INDUSTRY SAFETY DIRECTORATE
Government of India
Ministry of Petroleum & Natural Gas
8th Floor, OIDB Bhavan, Plot No. 2, Sector – 73, Noida – 201301 (U.P.)
Website: <https://www.oisd.gov.in> Tele: 0120-2593833

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PREFACE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January, 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (Mo&PNG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing inputs from national and international standards and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

**Executive Director
Oil Industry Safety Directorate**

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high-level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at **ps-ed.oisd@gov.in**.

CURRENT FUNCTIONAL COMMITTEE

NAME	DESIGNATION	ORGANIZATION
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S. Vinod Kumar	Group General Manager	ONGC
MEMBERS		
Upama Darshan (Till 31.03.24)	Chief General Manager	ONGC
J K Toppo	General Manager	ONGC
Sushant Roy	General Manager	ONGC
Kankusubhan Gogoi	Dy. General Manager	OIL
Biswajit Das	Dy. General Manager	OIL
Neeraj Shukla	Dy. General Manager	RIL
Amit Dwivedi	Sr. General Manager	RIL
Pramod Kumar	Astt. General Manager	EIL
Ashok Kumar Sana	Head Reliability	CAIRN
Bhanu Murty	Sup. Engineer	CAIRN
Bakul Gupta	Additional Director	OISD
MEMBER COORDINATOR		
Nandini Bakre	Additional Director	OISD

OISD also recognize the support & cooperation of several other experts from industry, who contributed to the preparation, review and finalization of this document.

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1.0 INTRODUCTION:

Safety in Electrical System is of paramount importance in the hydrocarbon industry due to presence of hazardous atmosphere. Electricity constitutes one of the major sources of ignition by arcs, sparks and hot surfaces. Electrical hazards may cause injuries and fatalities to personnel due to electric shock and burn besides equipment damage and property loss.

The primary objective of preparing this standard is to ensure electrical safety in onshore and offshore drilling and work over rigs.

2.0 SCOPE:

This standard lays down the minimum safety requirements for commissioning, operation, maintenance and repair of electrical equipment / systems on onshore and offshore drilling and workover rigs in exploration and production industry.

3.0 DEFINITIONS:

- (1) **Approved:** Approved by the Chief Electrical Inspector by a general or special order in writing and subject to such conditions as he may specify therein, wherever applicable.
- (2) **Authorized person:** A person with required competence duly designated / authorized by the management to carry out a specified job.
- (3) **Circuit:** an arrangement of conductor or conductors for conveying electricity and forming a system or a branch of a system and protected at the origin;
- (4) **Circuit breaker:** a mechanical switching device, capable of making, carrying and breaking currents under normal circuit conditions and also making, carrying for a specified duration and breaking currents under specified abnormal circuit condition;
- (5) **Competent person:** a person who is capable of identifying existing and predictable hazards associated with his area of activities and who has authorization to take prompt corrective measures to eliminate them;
- (6) **Cut-out:** means any device for automatically interrupting the flow of electricity through the conductor when the current increases above a pre-determined value, and shall also include fusible cut-out;
- (7) **Cable tray:** Mechanical support system that provide rigid structural system for Electrical cables;
- (8) **Dead:** means at or about earth potential and disconnected from any live system and is used only with reference to current carrying parts when these parts are not live;
- (9) **Designated Person:** a person whose name appears in the record maintained under sub-regulation (2) of regulation 3 of CEA Reg. 2023 by the agent of any company operating in an oil-field or the owner of a drilled well in an oil-field or a contractor.
- (10) **Earthing in onshore:** means connection of the exposed conductive and extraneous parts of an installation to the main earthing terminal of that installation or connection of neutral of transformer or generator or equipment to general mass of earth or earth bonded bar of that installation;
- (11) **Earthing in offshore:** Electric connections between conductive parts and local earth (in this case local earth, ship's hull or metallic structure).
- (12) **'Earthing arrangement or earthing system'** means all the electric connections and devices involved in the earthing of a system, an installation or equipment;
- (13) **Earth Electrode:** a metal plate, pipe or other conductor embedded in the soil that makes a direct contact to the general mass of the earth;
- (14) **Electrical Safety Officer:** an electrical engineer for ensuring observance of safety measures specified

under CEA regulations 2023 in organization for construction, operation and maintenance of electrical system.

- (15) **Emergency switchboard**-Switchgear and control gear assembly which is normally supplied by the main switchboard but, in the event of failure of the main electrical power supply system, is directly supplied by the emergency source of electrical power or the transitional source of emergency power and is intended to distribute and control electrical energy to the emergency services for all electrical consumers essential for the safety of the crew and the unit under emergency conditions
- (16) **Equipotential bonding**-Electrical connection putting various exposed conductive parts and extraneous conductive parts at a substantially equal potential.
- (17) **Exposed conductive part**- Conductive part which can be touched readily, and which is not normally alive, but which may become alive under fault conditions.
- (18) **Extraneous conductive part**- Conductive part not forming a part of the electrical installation and liable to propagate a potential, including earth potential.
- (19) **Flameproof (FLP) enclosure**: Means an enclosure in which the parts which can ignite an explosive atmosphere are placed and which can withstand the pressure developed during an internal explosion of an explosive mixture and which prevents the spread of explosion to the explosive atmosphere surrounding the enclosure;
- (20) **Flammable**: capable of being easily ignited, burning intensely or having a rapid rate of flame spread;
- (21) **Flash point**: The minimum temperature at which the liquid gives so much vapour that this vapour, when mixed with air, forms an ignitable mixture and gives a momentary flash on application of a small pilot flame under specified conditions of test;
- (22) **Hand-held portable apparatus**: an apparatus which is so designed as to be capable of being held in the hands and moved while connected to a supply of electricity; or having its own power supply, like battery etc.;
- (23) **Hazardous atmosphere**: an atmosphere containing any flammable gas in a concentration capable of ignition or containing noxious beyond permissible limits specified as per the OISD STD 113 or its revised version;
- (24) **Hazardous area**: an area where hazardous atmosphere exists or is likely to occur;
- (25) **Hull return system**- system in which insulated conductors are provided for connection to one pole or phase of the supply, the structure of the unit or other permanently earthed structure being used for effecting connections to the other pole or phase;
- (26) CEA Reg. 2023: Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023.
- (27) **Intrinsically safe**: as applied to apparatus or associated circuits shall denote that any sparking that may occur in normal working is incapable of causing explosion of inflammable gas or vapour;
- (28) **Installation Manager**: the person appointed by the owner of the Drilling / workover rig to be in-charge of and responsible for all operation and activities on or in connection with the rig;
- (29) **Linked switch**: a switch with all the poles mechanically linked so as to operate simultaneously;
- (30) **Live**: electrically charged;
- (31) **Metallic covering**: mechanically strong metal covering surrounding one or more conductors;
- (32) **Mobile offshore drilling unit (MODU)**: It is a vessel capable of engaging in drilling operations for the exploration for or exploitation of resources beneath the seabed such as liquid or gaseous hydrocarbons, sulphur or salt.
- (33) **Neutral conductor**: that conductor of a multi-wire system, the voltage of which is normally intermediate between the voltages of the other conductors of the system and shall also include return wire of the

single phase system;

- (34) **OIM** : Offshore Installation Manager is the person appointed by the owner of the offshore drilling / workover rig to be in-charge of and responsible for all operation and activities on or in connection with the rig.
- (35) **OMR, 2017**: The Oil Mines Regulation, 2017 as amended from time to time;
- (36) **Open sparking**: sparking which owing to the lack of adequate provisions for preventing the ignition of inflammable gas external to the apparatus would ignite such inflammable gas;
- (37) **Portable apparatus**: an apparatus which is so designed as to be capable of being moved while in operation;
- (38) **Street box**: a totally enclosed structure, either above or below ground containing apparatus for transforming, switching, controlling or otherwise regulating energy;
- (39) **Shall**: 'Shall' indicates a mandatory requirement;
- (40) **Should**: 'Should' indicates that the provision is recommended as a good engineering practice;
- (41) **May**: "May" are to be considered desirable or good practices which are encouraged to implement but not necessary.
- (42) **Switch Gear**: denote circuit breakers, cut outs and other apparatus used for the operation, regulation and control of circuits.
- (43) **Switch Boards**: mean an assembly including the switchgear for the control of electrical circuits, electrical connections and supporting frame.

PART-1

(ELECTRICAL SAFETY ON DRILLING AND WORKOVER RIG AT ONSHORE)

4.0 HAZARDOUS AREA CLASSIFICATION AND USE OF ELECTRICAL EQUIPMENT IN HAZARDOUS AREA:

- 4.1 Area Classification:** Area classification is a method of analyzing and classifying the environment where explosive gas atmospheres may occur so as to facilitate the proper selection and installation of apparatus to be used safely in that environment, taking into account gas groups and temperature classes. The classification of installations in hazardous areas according to applicable standards is essential for safety, regulatory compliance, risk management, operational continuity, and legal considerations.

Objective of classifying the installation into hazardous areas:

- To ensure that sources of ignition are segregated from sources of flammable gas or vapour.
- To assist in the appropriate location of air inlets and outlets for ventilation systems or combustion equipment
- To define the extent of flammable gas level from vents
- To ensure that lifesaving appliances, radioactive stores and emergency control points are located in non-hazardous areas,
- To determine the location of flammable gas/ toxic gas detectors
- To determine the maximum allowable surface temperature for particular areas

In order to determine the type of electrical equipment suitable for use in different conditions of hazardous atmosphere, the area shall be classified into three zones viz. Zone-0, Zone-1 and Zone-2.

Zone 0: An area in which a hazardous atmosphere is continuously present.

Zone 1: An area in which a hazardous atmosphere is likely to occur under normal operating condition.

Zone 2: An area in which a hazardous atmosphere is likely to occur only under abnormal operating conditions.

For further details for onshore drilling and workover rigs area classification refer Annexure-1. This is the minimum requirement for area classification.

Additionally, companies can do risk analysis and declare extended hazardous area. For zone categorization OISD-STD-113 can be referred.

4.2 The following shall be ensured on site:

- a) Approved key plan of the installation showing the different classes of hazardous zones, shown in distinct pattern is maintained.
- b) Different classes of zones are distinctly marked at actual places in the installation by suitable notice boards or other effective means; and
- c) Every person in the installation is made aware of the dangers associated with and precautions to be taken in different classified hazardous zones.

4.3 Use of electrical equipment in hazardous area.

- 4.3.1** No person shall use electrical appliance, equipment or machinery including lighting apparatus in zone "0" hazardous area.

- 4.3.2** The electrical appliance, equipment, machinery or other material that are used in zone "1" or zone "2" hazardous area of the installation shall be of a type and specification conforming to relevant Indian standard or an international standard adopted by the Bureau of Indian Standards through harmonization.

- i) All signaling and telecommunication, remote control and insulation tester circuits shall be so

constructed, installed, operated, protected and maintained as to be intrinsically safe;

- ii) All cables shall be so constructed, installed, operated and maintained as to prevent risk of open sparking;
- iii) All apparatus including portable and transportable apparatus shall have the enclosures conforming to the relevant standards; and
- iv) All electric lamps shall be enclosed in increased safety enclosure type “e” in Zone-2 and flameproof enclosures in Zone-1.

- 4.3.3** Electrical apparatus used in hazardous areas should be manufactured, tested, marked, and installed in accordance with Indian standard or international standards and certified by an independent testing recognized laboratory.
- 4.3.4** Records of certification / test reports conforming to relevant standards from an Indian Govt. Labs or NABL accredited labs or IECEx accredited labs or ATEX notified body which is not a part of manufacturer's facility shall be maintained.
- 4.3.5** Selection of electrical equipment with type of protection in classified area shall be as per **Annexure-4**.
- 4.3.6** The integrity of FLP equipment shall be ensured through visual inspection at least once in a year or every rig up whichever is earlier as detailed in OISD-STD-137. Inspection check sheet shall be maintained as per **Annexure-5**.
- 4.3.7** For FLP enclosures, only FLP double compression glands shall be used.
- 4.3.8** Equipment used in hazardous areas shall be checked and ensured as per OISD-RP-147 (Inspection and Safe practices during electrical installation).
- 4.3.9** Records of all electrical equipment installed in Hazardous area shall be maintained.

5.0 EARTHING & BONDING:

The following shall be earthed by means of double and distinct earth connections in accordance with the CEA Reg. 2023, IS 3043 & IS 7689 and IEC 60364:

- a) System neutral unless floating neutral system is used ^{Note1}
- b) All electrical equipment.
- c) Metallic non-current carrying parts of all electrical equipment.
- d) All enclosures, mounting structures, skids etc. e.g. Rig Mast and sub-structure, Diesel oil storage tanks, gas conditioning skids, power control rooms (PCRs), bunk houses (crew huts). Etc.
- e) Metallic Cable trays and racks, lighting mast.

Note1: In cases where an IT system is used for reasons of continuity of supply, an insulation monitoring device or suitable ground protection system shall be provided to indicate the occurrence of a first fault from a live part to exposed-conductive-parts to earth.

- 5.1 Cables shields and armored cables, if any, shall be earthed.
- 5.2 Lighting poles, minimum single earthing shall be provided.
- 5.3 All connections of earthing system shall have clean surfaces and free from paints, enamels, grease or other materials of poor conductivity.
- 5.4 Each pit location shall be identified by suitable marker; its surroundings must be kept neat & clean and easily accessible.
- 5.5 Earth resistance & continuity of connection shall be tested at each rig-up or every six months whichever is earlier and records thereof shall be maintained.
- 5.6 The double and distinct earthing connections of all diesel storage tanks shall be placed at the diagonally opposite extremities of the tanks.
- 5.7 Lighting fixtures and other LT equipment shall be earthed through the extra core provided in the cable for this purpose.

- 5.8 Two numbers earth conductors of flexible cable (6 mm Sq. braided copper wire) with sturdy crocodile clips near diesel tank (connected to the nearest electrode) shall be provided for earthing diesel tanker/ trucks during unloading of diesel in tank.
- 5.9 All conductors of an earthing system shall have conductivity at all parts and all joints 'at least equal to 50% of that of largest conductor used solely to supply the apparatus, a part of which it is desired to earth.
- 5.10 No switch, fuse or circuit breaker shall be inserted in any earth conductor.
- 5.11 In case of grounded neutral system, each of the transformer LT side neutral and generator neutrals, if any, shall be earthed by two distinct electrodes.
- 5.12 Ensure that all portable appliances are provided with 3-pin plug and socket connections and that the metal work of the apparatus is effectively earthed. RCBO/RCCB/ RCD shall be used with 30 mA residual current devices for connection.
- 5.13 Integrated earthing network may be maintained as equipotential bonding reduces risk of potential difference among systems viz. Cable trays. Piping, vessels etc.
- 5.14 All joints in pipelines, valves, storage tanks and associated facilities and equipment for petroleum shall be made electrically continuous by bonding or otherwise; the resistance value between each joint shall not exceed 1 ohm.
- 5.15 The lightning protection network shall be interconnected with main earth grid at minimum two locations below the ground.
- 5.16 Refer **Annexure-2** for Minimum size of earthing conductors in onshore.
- 5.17 The resistance value of an earthing system to general mass of the earth shall be as below:
- For the electrical system and equipment, a value that ensures the operation of the protective device in the electric circuit but not in excess of 4 ohms.
 - For lightning protection, the value of individual earth pit resistance shall be less than 10 ohms.
 - 7 ohms for storage tanks
 - 1 ohm for main earth grid and bonding connections between joints in pipelines & associated facilities
 - 2 ohms for each electrode to the general mass of the earth ^{Note}
- Note:
- 1) In areas with high soil resistivity (e.g. quartzite, granite, lateritic soil etc.), individual pit resistance value up to 5 ohm is acceptable, but grid resistance shall not be more than 1 ohm.
 - 2) In exceptional cases, where the individual pit resistance of less than 5 ohm is not achievable, number of rods/pipes/plates may be connected in parallel to achieve a combined pit resistance of 5 ohms. In this case also grid resistance shall not be more than 1 ohm.
 - 3) Typical diagrams of resistances to general mass of earth are provided at Annexure-V of OISD-STD-137
 - 4) The electrical earthing layout diagram with earth pit numbers need to be prepared, approved and should be kept available.
- 5.18 The earth fault current shall not be more than 750 milliampere in installations of voltage exceeding 250 V and up to 1100 V and the magnitude of the earth fault current shall be limited to these specified values by employing suitably designed, restricted neutral system of power supply including neutral-ground monitoring protection system.

6.0 CABLE LAYING AND CABLE CONNECTION:

All the above ground cables shall be laid in trays, duly covered. The underground cable shall be shrouded by suitable pipe and buried under earth at 30 cms or more. Care should be taken to avoid any transit damage to the trays/ cables.

6.1 Tray Placing:

- i) Ensure proper leveling of trays to avoid tripping hazard.

- ii) Place prefabricated steps or sandbags near cable tray along the walkway, as required.
- iii) All cable trays shall be bonded and connected to the earthing system.

6.2 Cable laying:

- i) During laying, pull the cables uniformly to avoid any twist or stress points. Properly align and dress the cables.
- ii) Check the cables for any deformation, deterioration in insulation and screen condition and replace the defective cables accordingly.
- iii) Adjust the extra length of cables inside the trays in coils in such a way that it can be extended without any difficulty, if required.
- iv) Route of the underground cables shall be marked.
- v) Power and control cables should not be mixed to avoid any interference issues during operation. Control cables should be separated from power cables.
- vi) Data cables should be laid in a separate hanger. Data cables should preferably be laid through metallic conduits to avoid any interference issues during operation.
- vii) Overhead cables should be properly supported to reduce mechanical stress on cables
- viii) Cable laying and termination checks shall be done as per OISD-STD-147.

6.3 Cables and Plugs / Sockets:

- i) Check the Insulation resistance (IR) value of cable insulation for cables as under
 - a) Each core to ground (for this, earth reference to be taken from a properly grounded system).
 - b) Between any two cores.
 - c) Every Core of cable to sheath / screen

Minimum IR values shall be 1 MEGA-OHM up to 650/1100V. IR value shall be measured as per OISD-STD-137 and record shall be maintained thereof.

- ii) No water / moisture shall be permitted in the plug / sockets.
 - i) Ensure the integrity of plug, sockets and their connections.
 - ii) Cables at both the ends shall be marked with appropriate numbers/tag (as in cable schedule diagrams).
 - iii) Ex. Certified cable glands shall be used in hazardous area.
 - iv) Ex. Certified plug & socket shall be used in hazardous area.
 - v) Plugs, sockets or receptacles shall be properly marked for identification.

7.0 OPERATION, MAINTENANCE AND REPAIR PRACTICES:

7.1 General:

- 7.1.1 Recommended practices of OEM of respective equipment shall be followed and records maintained thereof. Inspection shall be carried out and record shall be maintained as per Schedule XI of CEA Reg. 2023.
- 7.1.2 Pre-spud / Pre-workover electrical safety check list, as per **Annexure-3** shall be followed.
- 7.1.3 The following tools/ Instruments conforming to BIS specifications or equivalent shall be available:
 - a) Multi-meter;
 - b) Insulation tester
 - c) Earth tester;
 - d) Tong tester;
 - e) Electrical tool box containing various general tools like Insulated pliers, Insulated screw drivers etc;
 - f) Test lamp;(optional)
 - g) Discharge stick;
 - h) Double ladder;(A type insulated)
 - i) Lux meter;

- j) Cell Tester;(optional)
- k) Tachometer; (optical)
- l) Thermometer; (infrared)
- m) FLP Torch;
- n) Fuse puller
- o) Non-Contact voltage tester
- p) Phase sequence Meter(optional)
- q) Portable Gas Detector with multi / individual function
- r) LOTO Kit
- s) Non-sparking tool

Note: Intrinsically safe tools/instruments shall be used in Zone 1 & 2

- 7.1.4 Designated person shall supervise the electrical work. No person shall be allowed to carry out operations, work or testing on any part of the electrical system unless they meet the required competency and are authorized to perform the task. Competency requirements, training, validation, and authorization methodology should shall be prepared.
- 7.1.5 Other than mandatory MVT training as per MVT Rules, training as suggested in OISD-STD-176 for electrical persons should be ensured.
- 7.1.6 Temporary connections, naked joints & wiring shall be avoided. Inspection shall be done by designated person for unavoidable temporary connections prior to their use to prevent unsafe situations and record shall be maintained in the form of hot work permit.
- 7.1.7 On primary side of the transformer, a linked switch with fuses or circuit breaker of adequate capacity as per CEA Reg. 2023 shall be provided.
- 7.1.8 Electrical safety shoe shall be worn during electrical O&M activities. Electrical safety shoes shall be as per OISD-STD-155 and ISI marked as per IS 15298.
- 7.1.9 Suitable discharge rods shall be used to ensure that the electrical circuit is completely dead before commencing work on it. Non-Contact Voltage tester shall be used prior to discharge any circuit and ensure no voltage before discharging of the circuit.
- 7.1.10 Single pole switch or fuse in a neutral circuit shall not be connected.
- 7.1.11 Extra precautions shall be taken when working in damp areas.
- 7.1.12 Water and foam shall not be used for extinguishing electrical fire unless the equipment is de energized. Fire Extinguishers meant for electrical system (CO₂/ Class C fire extinguisher - Dry Powder / clean agent) shall only be used for electrical fires.
- 7.1.13 Firefighting facilities shall be provided as per OISD-STD-189 on "Fire Fighting Equipment for Drilling Rigs, Work Over Rigs and Production Installations".
- 7.1.14 Insulation resistance of motors and other electrical equipment shall be measured regularly with the help of suitable insulation tester once every six months or every rig-up whichever is earlier and records shall be maintained.
- 7.1.15 The effectiveness of switchgear and protective systems shall always be kept in working order and shall be checked by calibrating and testing at least once in a year and the result thereof shall be recorded in separate register kept for the purpose and in electronic form. However, trip trial can be taken in every rig up for switchgear & protective device.

Wherever numerical relays are being used they shall be checked by testing procedure as per guidelines of original equipment manufacturer or the relevant standards and periodicity of such checking shall be at least once in a year.

- 7.1.16 Functional testing of Earth Leakage Relays (ELR) and RCD (Residual Current Device) as per CEA Reg. 2023 shall be ensured once in a year or every rig up whichever is earlier. Records shall be maintained.
- 7.1.17 All interlocks shall be maintained in working conditions and inspected as per maintenance schedule. If any interlock is bypassed, then record shall be maintained in bypass register.
- 7.1.18 Any modification in the electrical equipment used in hazardous area shall be carried out in approved/ suitable manner. Management of change (MOC) procedure of OISD-STD-178 to be followed for any kind of change as per clause 4.0 of the standard or relevant clause of its latest version.

- 7.1.19 Standard and proper size/rated cables, glands, plugs, sockets, fuses, relays and circuit breakers shall be used.
- 7.1.20 All transformers, batteries, stationary & rotating electric equipment etc. including portable/ hand held electrical apparatus shall be inspected as specified in OISD-STD-137 on "Inspection of Electrical Equipment" and records maintained thereof.
- 7.1.21 All testing & measuring and tripping devices and instruments shall have valid calibration certificate.
- 7.1.22 During maintenance, if any equipment is removed from panel etc. which leaves an open cut-out of the equipment, the same, shall be promptly sealed. Blanking plates or other means shall be used to ensure that there is no loss of degree of ingress protection. It shall be ensured that all unused cable entries are blanked off without any loss of ingress protection.
- 7.1.23 Smoke detection with an audio/ visual alarm system shall be installed in PCR. The alarm system shall be duly tested once in a year or every rig up.
- 7.1.24 List of designated personnel to work on electrical systems shall be displayed in Power Control Room.

7.2 Isolation / energization of electrical equipment:

- 7.2.1 Work Permit System shall be followed as per OISD STD 105 on "Work Permit System". Detailing in respect of de-energization and energization shall also be available in the shift logbook. Log Sheet format for oil fields is given in Schedule XI of CEA Reg.2023. Electrical Isolation / energization permit format of OISD-STD-105 is given in **Annexure-6**. Alternatively, electrical isolation/ energization permit format of IS:5216 can be used provided it is interfaced with Hot/ Cold work permit of OISD-STD-105.

7.2.2 Every RIG shall have approved electrical isolation procedure for Motors, Transformers, Generators, VFD, Panels, UPS, TDS, lighting system, feeder, MCC, switchboard etc.

7.2.3 The electrical isolation procedure shall encompass the isolation and energization processes, including but not limited to the following items/systems as applicable:

- a) Circuit breaker / Isolator
- b) Interlocks
- c) Space heater and other auxiliaries
- d) Control supply
- e) Change over switch
- f) UPS
- g) Plug sockets and receptables
- h) Identification and mitigation of risk due to other nearby energy sources
- i) Ensure non availability of voltage in the circuit through non-contact voltage tester.
- j) Discharge all isolated equipment
- k) Check if additional grounding is required.
- l) LOTO procedure.
- m) Display of Caution boards.
- n) Work permit system.

7.2.4 Any LOTO meeting bypass requirement shall have applicable authorizations as per Bypass Policy.

7.3 Rig skidding operations:

- 7.3.1 Ensure sufficient length of power cables & control cables etc. is available before skidding.
- 7.3.2 Remove earth electrodes (if any) and cables falling in the path of dragging or working area.
- 7.3.3 Isolate power supply to mast lights and other electrical system not required in skidding operation.
- 7.3.4 Ensure that all electrical fittings on the Mast are secured.

7.3.5 Cables shall be checked for any pulling/ twisting after completion of skidding operation.

7.3.6 Prior to re-energization, ensure health of all electrical equipment connected with rig mast & substructures.

7.4 Rig down, transportation and rig up:

7.4.1 Power supply to all the electrical equipment not required during rig down operations shall be de-energized and LOTO is implemented.

7.4.2 A route survey shall be carried out to ensure minimum clearance from overhead lines of the equipment to be transported, as per CEA Reg. 2023. In case, minimum clearance requirement can not be met, then power lines shall be de-energized in coordination with State Electricity Board.

7.4.3 Ensure no HT/LT overhead transmission line passes near the drill / workover site.

7.4.4 The minimum clearance above ground and across road surface of National Highway or Expressway or State Highway or other road or highest traction conductor of railway corridor or navigational or non-navigational river of the lowest conductor of an alternating current overhead line, including service lines, of nominal voltage shall have the values specified in Schedule VIII A of CEA Reg. 2023.

7.4.5 Risk of electrocution of personnel associated with road transportation of rigs needs to be prevented through an appropriate system of checks and control measures.

7.4.6 Care shall be taken to avoid ingress of moisture to the electrical machines and components during transit and storage.

7.4.7 Cables to be properly secured during transportation.

7.4.8 Care shall be taken while handling and placing PCRs as described in OISD-GDN-218 on "Guidelines For Safe Rig- Up and Rig- Down Of Drilling and Workover Rigs".

7.5 Lighting System:

7.5.1 The voltage of lighting system shall be max. 250 volts phase to phase in accordance with CEA Reg. 2023. The neutral or mid points should also be connected to earth as per the requirements of CEA Reg. 2023 (Cl.104).

7.5.2 All connections to the light fitting in hazardous area shall be through double compression flameproof glands.

7.5.3 Each circuit shall be protected / controlled by ELR / RCD as per CEA Reg. 2023.

7.5.4 Aviation warning lights shall be fixed on top of the mast through a separate circuit.

7.5.5 Adequate illumination levels shall be ensured in all areas as per Annexure-7.

7.5.6 Emergency lighting shall be provided in PCR /emergency genset, mast, BOP control unit, rig floor, below derrick floor etc

7.5.7 An adequate number of portable hand lamps certified for use in hazardous area shall be kept available for immediate use in emergency and the voltage shall not exceed 30 V wherever applicable.

7.5.8 Every electrical lighting apparatus used in hazardous area of a rig shall be tested, marked, and installed in accordance with Indian standard or international standards and certified by an independent testing recognized laboratory.

7.6 Electrical Manpower/workforce:

7.6.1 Rig shall have designated Electrical Safety Officer as per CEA 2023 for ensuring electrical safety. Electrical Safety Officer shall carry out recommended periodic tests as per the relevant standards, and inspect such installations at intervals not exceeding one year and keep a record as per CEA Reg. 2023, Reg. 5.

7.6.2 The Electrical Safety Officer shall possess a degree in Electrical Engineering with at least 5 years' experience in operation and maintenance of electrical installation or a Diploma in Electrical Engineering with at least 10 years of experience in operation and maintenance of electrical installation.

7.6.3 One Electrical Supervisor in each shift of operation/ maintenance shall be appointed. Supervisor must have a valid Electrical Supervisor's Certificate of Competency, covering mining installations issued by the Appropriate Government authority.

7.6.4 One electrician in each shift of operation/ maintenance shall be appointed and every Technician must have a Wireman's license from State Board. The Maintenance in-charge/ IM must authorise the appointment of personnel.

7.6.5 The electrical supervisor shall be responsible for the proper performance of the following duties:

- a. thorough examination of all apparatus, including the testing of earth conductors and metallic coverings for continuity, as often as may be necessary to prevent danger; nullify potential risk and
- b. examination and testing of all new apparatus, and of all apparatus, re-erected in the ~~mine~~ rig before it is put into service in a new position.

7.6.6 In the absence of designated electrical supervisor, his substitute of equal competency shall be appointed in writing by Maintenance in-charge/ IM. The electrical supervisor or the substitute electrical supervisor appointed to replace him shall be personally responsible for the maintenance at the oil-field, of a log-book made up of the daily log sheets prepared in the form set out in Schedule XI of CEA Reg.2023 and the results of all tests carried out in accordance with the provisions of clause 7.6.5 shall be recorded in the log-sheets prepared in the form set out in Schedule XI.

8.0 WELL LOGGING OPERATIONS:

Following precautions related to electrical equipment / procedures during logging shall be followed:

- 8.1 Hold Safety meeting with Installation Manager, HSE Manager, Logging Team Head and Electrical supervisor.
- 8.2 All the arc/gas welding machines and cathodic protection equipment to be turned off.
- 8.3 Radio Transmitters/ Receivers, mobile phones and all generators to be turned off within 300 m. of work area well before arming the explosive device and shall remain switched off till the device is 70 m down inside the well.
- 8.4 Power supply to SRP (Sucker Rod Pump) within 30 m. from Well Head, shall be switched off.
- 8.5 Do not carry out any electrical testing (I.R. tester) discharging etc. during Well Logging operation.
- 8.6 AC power of the logging unit can be restored after running the device inside the well at least 100 m. below ground level.
- 8.7 Ensure before logging, static charge of the logging unit shall be earthed.
- 8.8 Verify that the voltage between the wellhead, rig, and logging unit does not exceed 0.25 volt.
- 8.9 If there is high tension line running over or underground within 30 meters of the work area, power must be disconnected. (Not applicable if, RF Safe RED detonators are being used, provided that any such line is not running over or underground from the area of explosive handling).
- 8.10 Ensure that all power supplies within the unit are properly grounded to prevent floating voltages or ground loops, which can damage the equipment. The grounding leads shall be connected to the ground of the logging unit, which shall be connected to the rig ground.
- 8.11 All metallic structure that might touch logging cable shall be grounded together to the logging unit.
- 8.12 During logging operation, permit system as per OISD STD 105 shall be followed.

For details, refer OISD-STD-183 on "Standard on Logging Operations"

9.0 CLUSTER WELL LOCATIONS:

During drilling / workover operation at cluster well locations, following shall be ensured:

- 9.1 Prior to switching on power supply, gas free atmosphere shall be ensured at the drill site.
- 9.2 Presence of hydrocarbon shall be monitored continuously during the operation. If concentration of inflammable gas exceeds twenty percent of its lower explosive limit a system should be in place to activate an audio alarm at appropriate locations. On activation of such alarm immediate action shall be taken to make operations safe and to isolate the cause in order to ensure safety of men, equipment, environment. In case the concentration rises to forty percent of lower explosive limit, the supply of electricity shall be cut-off immediately from all cables and apparatus lying within thirty metre of the installation and all sources

of ignition shall also be removed from the said area and normal work shall not be resumed unless the area is made gas-free.

- 9.3 All non-FLP motors or electrical equipment, within 15 m of the well to be mechanically scrapped, must be shut down and locked out for the duration of the scrapping activity.
- 9.4 No electrical cables shall pass over the producing well. All the cables shall run through the cable tray maintaining a minimum safe distance of 3 meters from the producing well.
- 9.5 During scraping operation, hot work permit system shall be followed as per OISD STD 105.
- 9.6 During scrapping, no electrical test shall be carried out using electrical test equipment unsuitable for hazardous area.

For details, refer OISD-STD-186 on "SIMOPS in E&P industry".

10.0 WELDING AND CUTTING OPERATIONS:

Common hazards involved in welding/cutting are sparks, molten metal, flying particles, harmful light rays, electric shocks, depletion in O₂ concentration due to generation of toxic gases, etc.

Following safety measures shall be implemented for Onshore Rigs:

- 10.1 Issue the hot work permit before starting the job and follow all conditions stipulated in the permit. Hot work permit shall be obtained before energization of welding equipment as per OISD STD 105. The person who has issued the permit shall explain the content of the hot work permit to the concerned welder and take his signature thereon to that effect before commencement of hot work.
- 10.2 Return cable should be connected as close to the welding point as far as possible.(Ref GDN-192).
- 10.3 Three core cable with proper colour code shall be used for connection of single phase welding machine (Ref. OISD-STD- 173 on "Fire Prevention and Protection System for Electrical Installations").
- 10.4 Work shall be carried out under supervision of a competent person.
- 10.5 Welder shall be imparted training about hazards associated with welding and electric shocks and nearby electrical equipment.
- 10.6 Ensure use of PPE like welding suits, leather gloves, safety shoes, safety goggles, helmets etc by the welder while working.
- 10.7 No person other than a competent welder duly authorized by the manager shall carry out welding or cutting work requiring use of flame or electric welding apparatus.
- 10.8 No work such as welding or cutting or grinding shall be undertaken in hazardous area unless the area is duly examined and found gas free by a competent person authorised for the purpose and a report of every such examination shall be recorded and maintained. Continuous gas monitoring shall be done during welding in hazardous area (zone1 & zone 2).
- 10.9 Welding machine shall be properly earthed and kept in a safer location and ventilated area.
- 10.10 Cables near the welding area shall be covered with fire blanket prior to commencement of the job.
- 10.11 If Welding Transformer is used, it shall be kept in safe area (outside hazardous area), and single length welding cable without any joint, shall be used.
- 10.12 RCD / residual current circuit breaker (RCCB) of suitable rating shall be installed on electrical circuits/ cables feeding power supply to the welding machine.
- 10.13 Appropriate Welding Booth to be made using fire blankets/ curtains, and shields to cover the work area and protect surrounding areas and workers.
- 10.14 Adequate number of suitable type of fire extinguishers and other firefighting arrangements (with firefighters), are made and kept readily available for immediate use, during the HOT work, and at least 30 minutes after the work is completed, or the work is in hot condition.
- 10.15 During any work of welding, cutting or grinding, the welder shall ensure that all flammable material, oil

grease, oil-soaked earth are removed from the area. No match, lighter, or smoking apparatus or any other source capable of igniting flammable gas is present at or around his place of work .

- 10.16 Workers shall be trained in emergency procedures and fire extinguishing techniques.
- 10.17 A fire-watcher in the work area shall be engaged for at least 30 minutes after work is completed and welding machine is de energized.
- 10.18 Welding area to be cordoned off and caution boards to be displayed. To prevent any unwarranted entry in the work area and also to caution other personnel taking actions which may endanger people working on the job, precautionary tags / boards are to be provided to display like "No Entry" sign on roads or "Caution - Men at Work Inside" etc.

Refer OISD-GDN-192 & IS 3016 for safe practices to be followed while cutting/ welding.

11.0 PRE- COMMISSIONING CHECKS:

- 11.1 Every Rig shall have approved Pre-commissioning check list for Motors, Transformers, Generators, PCR, BOP control unit, Eddy current brake, Drillers control, VFD, Panels, UPS TDS, lighting system, feeder, MCC, switchboard etc.
- 11.2 Inspect the electrical equipment for any transit damage and take corrective actions if necessary before start up.
- 11.3 The Pre-commissioning checks for electrical system of rig prescribed by OEM are to be carried out.

12.0 DOCUMENTATION:

Documentation shall be maintained as detailed out in Pre-spud / Pre-workover electrical safety check list as depicted in **Annexure-3**.

13.0 LIGHTNING PROTECTION:

- 13.1 For lightning protection to installations of oil mines fields, drilling rig, and workover rig the provisions as mentioned in IS/IEC 62305, OISD-GDN-STD-180 shall be followed.
- 13.2 The earthing system used in conjunction with LPS shall be in accordance with IS 3043 and be designed in such a manner as to prevent the risks of conduction, induction and feedback while discharging the lightning strike to earth.
- 13.3 A common integrated network for lightning protection, power systems & telecommunication systems may be preferred.
- 13.4 For lightning protection, the value of individual earth pit resistance shall be less than 10 ohms.
- 13.5 The remote earthing shall be adopted if the local grounding conditions are not favorable or in case of space constraint.
- 13.6 A protective system need not be fitted to a unit of metallic construction, where a low resistance path to earth will be inherently provided by bolted and welded steelwork from the highest point of the unit to earth.
- 13.7 Metallic masts and metallic structural members may form part or all of any protective system. If the metallic structure/steel structure thickness is greater than 4.8 mm thickness, there is no additional lightning protection (i.e. air termination & down-conductor) is required. However it is desirable to have LPS as per lightning system study.
- 13.8 It is very important that the down conductors shall follow the most direct path possible between the air termination and the earth termination, avoiding sharp bends, upturns and kinks. Joints shall as far as possible be avoided in down conductors. The formation of loops on the down conductor shall be avoided or minimum separation distance(s) shall be maintained. Adequate protection may be provided to the conductors against mechanical damage. In deciding on the routing of the down conductor, its accessibility for inspection, testing and maintenance should be taken into account.(IEC 62305 Cl5.3).

- 13.9 Earth Terminations – Each down conductor shall have an independent earth termination. It should be capable of isolation for testing purposes.
- 13.10 Earth Electrodes – Earth electrodes shall be constructed and installed in accordance with IS/IEC 62305-3. Type A or B earth electrode arrangement as per site conditions shall be applied.
- 13.11 All lightning protective systems shall be examined by a competent person after completion, alteration or extensions, in order to verify that they are in accordance with the recommendations of the Standard. A routine inspection shall be carried out at least once a year or every rig up whichever is earlier.
- 13.12 In every oil and gas drilling rig and workover rig a comprehensive Lightning risk assessment study shall be carried out as elaborated in IS/IEC-62305. For new geographical area having different climatic conditions, risk assessment shall be done again.

14.0 PROTECTION AGAINST ELECTRIC SHOCK:

- 14.1 Electrical hand tools shall be double-insulated or grounded in accordance with NFPA 70. Ground fault circuit interruption protection is recommended.
- 14.2 Every person who is working on an electric supply line or apparatus or both shall be provided with,
- personal protective equipment, tools and devices such as rubber gloves and safety footwear suitable for working voltage, safety belts for working at height, nonconductive ladder, earthing devices of appropriate class, helmet, non-contact line tester, hand lines, voltage detector and hand tools as per the relevant standards.
 - Any other device for protecting him from mechanical and electrical injury due to arc flash and such personal protective equipment, tools and devices shall conform to the relevant standards and shall always be maintained in sound working condition.
- 14.3 Employees shall use insulated tools or handling equipment, or both, when working inside the restricted approach boundary of exposed energized electrical conductors or circuit parts where tools or handling equipment might make accidental contact.
- 14.4 An approved insulated rescue tool (shepherd's hook) available in all switchgear / transformer rooms for the purpose of removing persons from live conductors. The hook must be inspected and stored as per the manufacturer recommendation.
- 14.5 If testing of MV systems / equipment is required, it must be performed with:
- Use of a non-contact line tester.
 - Appropriately rated equipment.
- 14.6 All maintenance personnel, emergency response team members and any person acting as a stand-by person for personnel performing electrical work must be trained in the use of the insulated rescue tool.
- 14.7 Guarding of Energized Conductors and Circuit Parts: Enclosures shall be maintained to guard against accidental contact with energized conductors and circuit parts and other electrical hazards. Covers and doors shall be in place with all associated fasteners and latches secured.
- 14.8 Warning signs, where required, shall be visible, securely attached, and maintained in legible condition on panels.
- 14.9 Personal safety and protective equipment such as the following shall be maintained in a safe working condition and inspection records to be maintained:
- 1) Grounding equipment
 - 2) Rubber gloves, sleeves, and leather protectors
 - 3) Insulating mats and similar insulating equipment
 - 4) Portable lighting units
 - 5) Temporary protective grounding equipment

- 6) Dielectric footwear
 - 7) Non-contact voltage line testers
 - 8) Insulated and insulating hand tools
 - 9) Arc Flash Protective Cloths (desirable)
 - 10) Rescue Hook (for electrical rig)
 - 11) Ambu Bag
 - 12) Face shield (while handling Battery)
 - 13) PVC Suit & gloves for handling acid/ KOH (while handling Battery).
 - 14) Gel blanket (desirable)
- 14.10 Every installation of voltage exceeding 250 V shall affix permanently in a conspicuous position a danger notices in Hindi or English and the local language of the district, with a sign of skull and bones of a design as per relevant standards on, - every motor, generator, transformer and other electrical plant and equipment together with apparatus used for controlling or regulating the same.
- 14.11 Effects of Electricity on the Human Body. The employer and employees shall be aware of the following hazards associated with electrical equipment. Effects of Power Frequency Current:
- a) At 0.5 mA, shock is perceptible.
 - b) At 10 mA, a person may not be able to voluntarily let go of an energized electrical conductor or circuit part.
 - c) At about 40 mA, the shock, if lasting for 1 second or longer, can be fatal due to ventricular fibrillation.
 - d) Further increasing current leads to burns and cardiac arrest.
- For further details refer NFPA 70 Annexure -1 for effects of electrical current in human body.
- 14.12 Measures shall be taken in stationary battery installations for protection against direct contact and indirect contact or both.

15.0 ELECTRICAL POWER SOURCE:

- i) Electrical power generating sets are to be placed in unclassified locations.
- ii) The Main electrical power source of all installations shall be of suitable capacity to be able to handle the maximum load requirement in normal operational condition.
- iii) Rig shall be equipped with an emergency source of electrical power, which shall be self-contained. The capacity of emergency power source shall not be counted towards maintaining normal operations.
- iv) Services essential for maintaining safety shall be ensured in the event of failure of the main power source.

15.1 EXHAUST OUTLETS:

Engine exhaust outlets should be located in unclassified locations whenever possible to minimize the risk of ignition of flammable mixtures. Stack height shall be as per OISD-STD-201 from occupational health and environment point of view. Engine exhaust outlets shall be fitted with spark arrester.

15.2 AIR INTAKES

It is recommended that engine combustion air intakes be located in unclassified locations whenever possible to minimize the risk of ingestion of flammable mixtures.

15.3 ENGINE-GENERATOR PROTECTIONS:

Appropriate controls should be provided to shut down all reciprocating engines that are driving generators and open the generator breakers when any of the following conditions occur:

- a. Low lube oil pressure.
- b. High jacket water temperature.
- c. Overspeed. Overspeed shutdowns should operate independently of governor controllers and should be set at no more than 115% of rated speed.
- d. Overvoltage for generators 500 kW and larger. It is recommended that either overvoltage shutdown controls be provided or that breakers be tripped and voltage regulators be de-energized.
- e. Overload and Short Circuit. The overcurrent trip setting should not exceed 115% of the generator full load current.

When differential protection is not provided, it is recommended that instantaneous breaker trips be used on generators that normally operate in parallel with two or more other generators. To ensure proper coordination the instantaneous trip setting should be above, but as close as practicable to, the maximum asymmetrical short circuit current available from any one generator.

- f. Reverse Power. When two or more generators are to operate continuously in parallel, each unit should be provided with a reverse power relay to trip the generator breakers in the event of reverse power flow.
- g. Undervoltage and Overvoltage. Undervoltage and overvoltage sensing devices with time delay trips should be considered for protection of the electrical system. An undervoltage trip device should open the generator main circuit breaker when the prime mover is shut down.
- h. Underfrequency and Over-frequency. Underfrequency and over-frequency sensing devices with time delay trips should be considered for protection of electrical systems.

15.4 SYNCHRONIZING CONTROLS:

It is recommended that the controls of generators intended to be paralleled be equipped with:

- a. Synchroscopes or synchronizing lights, or both, to show when generators are in phase. A synchroscope provides more accurate indication of phase relationship and should be considered in most applications for smoother switching operations. The synchronizing indicators should be visible from the speed and voltage setting controls.
- b. A synchronizing relay in the breaker closing circuit of electrically operated circuit breakers to prevent out-of-phase paralleling. Consideration should be given to the installation of automatic synchronizing controls on units greater than 250 kW.
- c. Interlocking controls to assure that all other generator circuit breakers for nonoperating generators and incoming feeders are open when an oncoming generator breaker is closed on a dead bus.

15.5 GROUND-FAULT DETECTION

- a. When the electrical system is ungrounded, a ground-fault indication system is recommended.
- b. When the electrical system is high resistance grounded, a ground-fault alarm is recommended.
- c. When the electrical system is low resistance grounded, ground fault protective devices should be provided to open the generator breaker if coordinated downstream devices do not clear the fault.
- d. When the electrical system is solidly grounded and the main generator protective device is rated 1000 amperes or greater, ground fault protective devices should be provided to open the generator breaker if coordinated downstream devices do not clear the fault. Consideration should be given to providing ground fault protection for generators with protective devices rated less than 1000 amperes.

15.6 METERING:

Minimum metering should include an ammeter (with a selector switch to meter all phases), a voltmeter, and a frequency meter. A voltmeter selector switch (to provide metering of all phases), a running time meter, a power factor meter, and a watt meter are optional.

16.0 EMERGENCY SHUTDOWN SYSTEM:

- a. Suitable emergency shutdown system shall be provided at the driller's controls, so that when concentration of inflammable gases exceeds forty percent of lowest explosive limit, the supply of electric energy from all generating sets shall be cut off immediately to all cables and apparatus lying within thirty metres of the installation and all sources of ignition shall be removed from the said area and normal work shall not be resumed unless the area is made gas-free or normalized.
- b. Provided that such disconnection shall not apply to intrinsically safe environment monitoring scientific instruments. Any such disconnection or reconnection of the supply shall be noted in the log sheet in hard copy and electronic form which shall be maintained in the form set out in Schedule XI of CEA Reg. 2023.
- c. Site specific emergency shutdown system shall be available and shall be operational. It should be tested once in a year or every rig up, whichever is earlier.

17.0 ABBREVIATIONS:

1. AC PCR: AC Power Control Room
2. ATEX: "ATmosphères EXplosibles," which translates to "explosive atmospheres",
3. BOP: Blow Out Preventer
4. CEA Reg. 2023: Central Electricity Authority Regulations 2023
5. DGMS: Directorate General of Mines Safety
6. DC PCR: DC Power Control Room
7. Ex: Explosion-proof
8. ELR: Earth Leakage Relay
9. FLP: Flameproof
10. IM: Installation Manager
11. IECEx: The certification by the International Electrotechnical Commission for Explosive Atmospheres.
12. IR: Insulation Resistance
13. LOTO: Lockout-Tagout
14. MODU: Mobile Offshore Drilling Unit
15. MVT: Mines Vocational Training
16. MOC: Management of Change
17. MCC: Motor Control Center
18. NABL: National Accreditation Board for Testing and Calibration Laboratories.
19. NFPA: National Fire Protection Association
20. OIM: Offshore Installation Manager
21. OMR: Oil Mines Regulation
22. RCD: Residual Current Device
23. RCBO: Residual Current Circuit Breaker with Overcurrent Protection
24. RCCB: Residual Current Circuit Breaker
25. RF: Radio frequency
26. RED: Rig Environment Detonator
27. SRP: Sucker Rod Pump
28. TDS: Top Drive System
29. UPS: Uninterrupted Power Supply
30. VFD: Variable Frequency Drive

18.0 REFERENCES:

CEA Regulations 2023, Oil Mines Regulations 2017, API RP 500-B,
IS 3043, 15652, 60079-1, 8280, 16724, 60079-2, 60079-11, 60079-15, 60079-7, 8985, 5572, 4051,
60034-5/2147, 60079-5, 11333, 3646, 13774, 7689, IEC 62305, IEC 60903, IS 15298 (part 2) ISO: 20345,
NFPA 70.
OSHAS standard – 1910.302-308
OISD STDs-105, 137,147, 149,173, 174, 155, 178, 180, 183, 186, 189, 192, 207,201 and 218.
Recommended Code of practices of ONGC, OIL, RIL, CAIRN.

19.0 ANNEXURES

Annexure-1

HAZARDOUS AREA CLASSIFICATION FOR DRILLING and WORK-OVER OPERATIONS:

(1) Well – head area:

- i) When the derrick is not enclosed and the sub-structure is open to ventilation, the area in all directions from the base of rotary table extending up to 3.0 metre shall be zone “2” hazardous area. Any cellars, trenches and pits below the ground level shall be zone “1” hazardous area; the area lying up to 3.0 metre in horizontal direction from the edge of any cellars, trenches on pit and 0.5 metre vertically above the cellars, trenches on pit shall be zone “2” hazardous area.
- ii) When the derrick floor and substructure are enclosed, the enclosed substructure below the derrick floor, including collars, pits or sumps below the ground level, shall be zone “1” hazardous area; the enclosed area above the derrick floor shall be zone “2” hazardous area.

(2) Mud Tank and Channel:

The free space above the level of mud in tank and channel shall be zone “1” hazardous area; the area in a radius of 3.0 metre in all direction from the edge of mud tank and channel shall be zone “2” hazardous area.

(3) Shale Shaker:

- i) The area within a radius of 1.5 metre in all directions from the shale shaker in open air shall be zone “1” hazardous area. The area beyond 1.5 metre and up to 3.0 metre in all directions from the shale shaker shall be zone “2” hazardous area.
- ii) When the shale shaker is located in an enclosure, the enclosed area shall be zone “1” hazardous area to the extent of the enclosure. The area outside the shale shaker and up to 1.5 metre in all directions from the shale shaker shall be zone “2” hazardous area.

(4) Degasser:

The area within a radius of 1.5 metre from the open end of the vent extending in all directions shall be zone “1”; the area beyond 1.5 metre and up to 3 metre in all directions from the open end of vent shall be zone “2” hazardous area.

(5) Desander and Desilter:

The area within a radius of 1.5 metre in all direction from the Desander and Desilter location in open air shall be zone “2” hazardous area.

(6) Effluent Pit and Open sump:

The free space above the level of flammable liquid within the effluent pit on sump shall be zone “1” hazardous area; the free space lying up to 3.0 metre in hazardous direction from the edge of any effluent pit or sump and 5.0 metre vertically above the effluent pit or open sump shall be zone “2” hazardous area.

(7) Gas conditioning skid : The area in a radius of 3.0 metre in all direction from the Gas conditioning skid shall be zone “2” hazardous area. (In new rigs)

(8) Diesel tank : The area in a radius of 3.0 metre in all direction from the Diesel tank shall be zone “2” hazardous area.

Also the following areas shall be known as hazardous areas

- i) an area of not less than ninety metre around an oil-well where a blow-out has occurred or is likely to occur, as may be designated by the Installation Manager or the senior most official present at the site;
- ii) an area within sixteen metre of an open discharge of petroleum bearing fluid from a well under production test;
- iii) an area within fifteen metre of:
 - a) a producing well-head or any point of open discharge of the crude there from or other point where emission of hazardous atmosphere is normally likely to arise; or
 - b) any wildcat or exploration well-head being drilled in an area where abnormal pressure conditions are known to exist;

iv) any area within three metre of:

- a) any producing well-head where a closed system of production is employed such as to prevent the emission or accumulation in the area in normal circumstances of a hazardous atmosphere; or
- b) exploration or interspaced well-head being drilled in an area where the pressure conditions are normal and where the system of drilling employed includes adequate measures for the prevention in normal circumstances of emission or accumulation within the area of a hazardous atmosphere;

Annexure-2

MINIMUM SIZE OF EARTHING CONDUCTORS

	<u>Type of Equipment/System</u>	<u>Type and Size of earthing conductors</u>
a)	Lighting poles	Through 3 rd core of the cable or 8 SWG GI wire
b)	Motor upto 3.7 KW and all PBS and instrumentation boxes	8 SWG GI wire
c)	Cable trays (Loop connection)	8 SWG GI wire
c)	Motors above 3.7 KW upto 30 KW and generators/alternators including welding receptacle	25 mm x 4 mm GI strip
d)	Motors and Generators/Alternators above 30 KW upto 100 KW	40mm x4mm GI Strip
e)	Generators/Alternators/DC Motors above 100 KW	50mmx4mm GI strip
f)	All Metallic Structures	40mmx4mm GI strip
g)	Power Transformer (Body)	50mmx4mm strip
h)	PCR	50mmx4mm strip

Note:

- 1) All earthing connections between the earth electrodes by GI strip of size 50 mm x 4 mm. However it should be based on calculation and higher size, if required, shall be provided.
- 2) In all above cases equivalent multi strand cable can also be used.

Annexure-3

PRE-SPUD / PRE-WORKOVER ELECTRICAL SAFETY CHECK LIST

The checklist should include but not limited to the following:

1. DISPLAY OF CAUTION BOARDS:

- a) Electrical Caution Board on all electrical equipment.
“DO NOT OPERATE, MEN WORKING ONLINE” wherever applicable.
- b) Notice forbidding entry of unauthorized persons in Electrical PCR / Switch Board Room. “ENTRY OF UNAUTHORISED PERSONS STRICTLY PROHIBITED”.

2. DISPLAY AT STRATEGIC LOCATIONS:

- i) Power flow diagram.
- ii) Cable Lay out and Earthing Drawing.
- iii) Hazardous area classification of the location.
- iv) Electrical Shock Treatment Chart.
- v) Fire Extinguishers layout.
- vi) Do's & Don'ts.
- vii) Role and responsibility of electrical personnel during emergency.
- viii) Display list of electrical authorized persons in PCR.
- ix) Important/emergency telephone numbers.

(Address and contact number of the nearest Doctor, Hospital with a facility for first-aid treatment for electric shock and burns, ambulance service and fire service at PCR and IM room.)

3. RECORD KEEPING :

- i) Employee's Bio- Data.
- ii) Shift Roster.
- iii) List of trained MVT, First Aid & Fire fighting personnels.
- iv) List of persons with Electrical License.
- v) Accident and Near-miss Records.
- vi) Equipment Logbook & maintenance record.
- vii) Commissioning Record.
- viii) Earth Resistance & Insulation Resistance Record.
- ix) Log sheet as per sub-regulation (9) of regulation 112, and Schedule XI as per sub-regulation (9) of regulation (117)] CEA Reg. 2023).
- x) Work Permits and isolation/energization records.
- xi) Daily check list of electrical equipment.
- xii) Periodical check list (including R/up & R/down).
- xiii) Record of all electrical equipment installed in Hazardous area
- xiv) Inspection records of FLP equipment.
- xv) Inspection record of tools/instruments as listed in CI 7.1(iii)

- xvi) Measuring devices/Instrument calibration certificate.
- xvii) List of PPEs
- xviii) Records of Functional testing of Earth Leakage Relays (ELR) and RCD
- xix) Records of Illumination level
- xx) Form IV of CEA Reg. 2023.

4. EARTHING:

- i) Total No. of earth electrodes /pits.
- ii) Condition of earth electrodes, G.I. strips & earthen wires.
- iii) 2 Separate & independent earthing exists for all electrical equipment.
- iv) 2 Separate & independent earthing exists for substructure for lightning protection.
- v) Value of earth resistance.
- vi) Equipment, cable tray & C.G.I sheet roof etc. connected to earthing system – Yes / No.
- vii) 2 Separate & independent earthing exists for Transformer body.
- viii) 2 Separate & independent earthing exists for neutral.

5. PROTECTIVE EQUIPMENT:

- i) Emergency shutdown system at Drillers console exist & functional.
- ii) Automatic disconnection facility applicable for drilling motors/main generators and other motors under fault condition is available.

6. ELECTRICAL EQUIPMENTS FOR HAZARDOUS AREA APPLICATION:

- i) Condition of FLP equipment.
- ii) Any modification in the electrical equipment used in hazardous area carried out in approved/suitable manner.

7. VOLTAGE RESTRICTION:

230 volts Phase to Phase and 110 Volts Phase to Earth voltage system for lighting.

8. USE OF PERSONAL PROTECTIVE EQUIPMENT:

- i) Insulating Hand Glove (As per IS 13774) The maximum use voltage recommended for each class of glove is designated in the following table.

Class (Colour code of symbol)	AC RMS (V, r.m.s)	DC (V)
00 (beige)	500	750
0 (red)	1000	1500
1 (white)	7500	11500
2 (yellow)	17000	25500
3 (green)	26500	39750
4 (orange)	36000	54000

The maximum use voltage is the rated a.c. voltage r.m.s. of the protective equipment that designates the maximum nominal voltage of the energized system that may be safely worked.

- ii) Insulating mat as per IS 15652(2006).
- i) Insulated Shoes.
- ii) Coverall, Helmet, safety belt, goggles, masks etc.
- iii) Arc suit
- iv) Rescue Rod
- v) SCBA (Self Contained Breathing Apparatus)

9. Aviation warning light fittings at the top of the mast.

10. Ensure sufficient illumination level as per CI 7.5.

11. Availability of the following at site as applicable:

- i) Double sided ladder.
- ii) First Aid box and stretcher.
- iii) FLP torch.
- iv) Standby / emergency generator set.
- v) Emergency lighting inside Control Room.

Annexure-4

Type of Protection for Electrical Equipment for Classified Areas

- i) The basis for hazardous area classification recognizes the differing degrees of probability with which flammable atmosphere may arise in the installation, in terms of the frequency of occurrence and the probable duration of existence on each occasion.
- ii) Following factors shall be considered for proper selection of electrical apparatus and equipment for areas where flammable gas or vapour risks may arise:
 - a) Area classification, i.e. Zone 0, 1 or 2
 - b) Gas group classification, i.e. gas groups IIA, IIB or IIC.
 - c) Temperature classification i.e. T-Rating.
 - d) Environmental conditions in which apparatus is to be installed.

Wherever practicable, electrical apparatus in general and switch and control apparatus in particular shall be installed in safe area. PCR shall be located in safe area.

General guidelines for type of protection for electrical equipment in hazardous areas are enumerated as Table-I:

Table-I: TYPES OF PROTECTION

Area Classification	Description	Symbol
Zone 0 (See Note 1)	Intrinsic safety category 'ia'	'ia'
	Encapsulation	'ma'
	Other electrical apparatus, specifically designed for Zone 0.	's'
Zone 1	Any type of protection adequate for Zone 0.	
	Intrinsic safety category 'ib'	'ib'
	Flame proof enclosure (See Note 6)	'd'
	Pressurization (See Table II)	'p', 'px', 'py'
	Powder filling	'q'
	Encapsulation	'm', 'ma', 'mb'
	Other electrical apparatus specifically designed for Zone 1	's'
	Oil Immersion (see note 4)	'o'
Zone 2	Any type of protection adequate for Zone 0 or Zone 1	
	Intrinsic safety category 'ic'	'ic'
	Increased safety (See Note 2)	'e'
	Non-sparking apparatus (See Note 2)	'n'
	Pressurization (See table II)	'pz'
	Encapsulation	'm', 'ma', 'mb'

Note 1 It may be noted that as per petroleum rules the use of intrinsically safe electricity apparatus in zone '0' area is permitted only when the use of such apparatus cannot be completely excluded, whereas the Oil Mines regulations prohibit use of any electrical equipment in zone '0' areas.

Note 2 The apparatus with type of protection 'e' and type of protection 'n' when installed outdoors shall have enclosures having the ingress protection as recommended in IS/IEC-60529.

- a. IP 55 where there are uninsulated conducting parts internally, and
- b. IP 44 for insulated parts.

In case of usage of Ex-nA type of motors the requirement of pre-start purging shall be verified based on requirements of IEC-60079-15.

Note 3 Name plate details of equipment intended for use in hazardous area shall include relevant marking of hazardous protection as per applicable codes.

Note 4 Oil-immersed apparatus may be used only in case its security will not be impaired by tilting or vibration of the apparatus.

Note 5 Flameproof enclosures are covered under compulsory certification.

Note 6 In Zone '1' areas, Ex d/e equipment i.e. Ex 'd' equipment with Ex 'e' terminal boxes is also permissible.

Table-II

**MINIMUM ACTIONS ON FAILURE OF PROTECTIVE GAS FOR TYPE OF PROTECTION
'P'**

Area Classification	Enclosure does not contain ignition capable apparatus	Enclosure contains ignition capable apparatus
Zone 1	Alarm	Alarm and switch off
Zone 2	No action required	Alarm

Annexure-5

Periodic Visual Inspection check list of FLP equipment (as per IEC 60079)

Sl. No.	Description	Observation
1.	Is there any visible un-authorized modifications.	
2.	Bolts / screws / other fasteners are tight and complete.	
3.	All blanking elements, glands are of correct type, complete, in good condition and tight.	
4.	Is there any unused cable entry left open?	
5.	Is there any visible/ obvious damage to cables?	
6.	Is there any obvious crack / damage to enclosure.	
7.	Are there any obstructions present near flanged joints?	
8.	Sealing of trunking, ducts, pipes and /or conduits is satisfactory.	
9.	Is there any undue accumulation of dust and dirt?	
10.	Earthing and bonding connection are in good condition and tight.	
11.	Enclosure glasses, sealing gaskets and / or compounds are satisfactory (water tightness).	
12.	Is there any damage to window glasses /transparent covers?	
13.	Ducts, pipes and enclosures are in good condition.	
14.	Circuit identification present and correct. (applicable for all controls and major equipment)	
15.	Is there any undue corrosion.	
16.	Is there any undue vibration.	
17.	Name plate is in place and readable.	
18.	The pressure relief plate (metallic / nonmetallic) provided in the cable box is intact.	
19.	Is the cable box unusually hot.	
20.	Are all relevant updated drawings and documents available	
21.	Earth continuity / bonding connections are intact.	

**Annexure-6****TYPICAL ELECTRICAL ISOLATION/ ENERGIZATION PERMIT**

(NAME OF ORGANIZATION)

(LOCATION)

SECTION-A: ISOLATION PERMIT

Permit no.: _____ Cross-Reference of other Permit: _____

Request for Isolation:

Date: _____ Time: _____

Department/ Section/ Area issuing the permit _____

Equipment number to be isolated: _____

Name of the equipment/ circuit to be isolated: _____

Description of the work to be carried out: _____

The above-mentioned equipment/ circuit shall be de-energized and isolated from all live conductors to carry out the maintenance work by _____ section/ for operational requirement.

Issuer Name**Designation****Signature****Certificate of Isolation:**

Date: _____ Time: _____

Certified that Equipment/ Circuit no. _____ of _____ plant has been made dead, is electrically isolated from all live conductors and has been connected to earth and the work mentioned above can now be carried out in accordance with the safety rules and regulations. For the purpose of making the equipment dead, the switches/ isolators/ links/ fuses (tick as applicable), have been opened and the section so isolated has been earthed at each isolation point and danger notice plates fixed thereon. The LOTO tag/ device no. _____ is put on the supply panel. Actions in respect of electrical isolation have been recorded in the electrical shift logbook including the instructions for the person who may relieve me.

Name of Authorized Person**Designation****Signature**

**SECTION-B: ENERGIZATION PERMIT**

Permit no.: _____

Request for Energization:

Date: _____ Time: _____

Department/ Section/ Area issuing the permit _____

Equipment number to be energized: _____

Name of the equipment/ circuit to be energized: _____

The above mentioned work on the subject equipment/ circuit has been completed and all the applicable work permits are closed. This equipment/ circuit may be energized.

Issuer Name_____
Designation_____
Signature**Certificate of Energization:**

Date: _____ Time: _____

Certified that Equipment / circuit no. _____ of _____ plant has been electrically energized by closing the switches/ isolators/ links/ fuses (tick as applicable), and the equipment put back into service.

The danger LOTO tag/device no. _____ is removed from the supply panel. The temporary earths and other connections and danger notice plates made have been removed. Necessary entry is also recorded in the electrical shift logbook.

Name of Authorized Person_____
Designation_____
Signature

Annexure-7

ILLUMINATION LEVELS AT D/RIG AND WOR

Area	Illumination level in Lux
Open machinery area	70-100
Derrick floor	100-200
Pipe Rack	70-100
Monkey Board	70-100
Power Control Room	200-300
Diesel tank Area and transfer pump	70 - 100
Compressor house	100-200
Chemical loading area	70-100
Mud tank and mud return channel area	70-100
Sub structure	100-150
Engine room	70-100
Security lighting and peripheral area	50 - 100
Staircase	50 - 100
Fire Pump Area	50 - 100

PART-2

(ELECTRICAL SAFETY ON DRILLING AND WORKOVER RIG AT OFFSHORE)

4.0 Hazardous area classification and use of electrical equipment in hazardous area:

4.1 Area classification:

Area classification is a method of analyzing and classifying the environment where explosive gas atmospheres may occur so as to facilitate the proper selection and installation of apparatus to be used safely in that environment, taking into account gas groups and temperature classes. The classification of installations in hazardous areas according to applicable standards is essential for safety, regulatory compliance, risk management, operational continuity, and legal considerations.

4.2 Objective of classifying the installation into hazardous areas:

- To mitigate the risk of fire and explosion by appropriately designing and selecting electrical equipment and wiring methods suitable for the specific hazardous conditions present.
- To ensure that sources of ignition are segregated from sources of flammable gas or vapour.
- To assist in the appropriate location of air inlets and outlets for ventilation systems or combustion equipment
- To define the extent of flammable gas level from vents.
- To ensure that lifesaving appliances, radioactive stores and emergency control points are located in non-hazardous areas
- To determine the location of flammable gas/ toxic gas detectors
- To determine the maximum allowable surface temperature for particular areas

4.3 In order to determine the type of electrical equipment suitable for use in different conditions of hazardous atmosphere,

The area shall be classified into three zones viz. Zone-0, Zone-1 and Zone-2.

- Zone 0:** An area in which an explosive atmosphere is continuously present or present for longer period.
- Zone 1:** An area in which a hazardous atmosphere is likely to occur under normal operating condition.
- Zone 2:** An area in which an explosive gas atmosphere is not likely to occur in normal operation, but, if it does occur, will exist for a short period only.

For the area classified as per class / division.

- Class I location** – A location in which flammable gases or vapours, as may be, present in the air in quantities sufficient to produce explosive ignitable mixture.
- Class I Division I location:** A location in which ignitable concentrations of flammable gases or vapour are expected to exist under normal conditions or in which faulty operation of equipment or process might simultaneously release flammable gases or vapours and might also cause failure of electric equipment.
- Class I Division II location:** A location in which flammable gases or vapour may be present but

normally are confined within closed systems, are prevented from accumulating by adequate ventilation, or the location is adjacent to a Division I location from which ignitable concentrations might occasionally be communicated.

Refer **Annexure-1** for Hazardous area classification and extent of classified locations.

4.4 The following shall be ensured in Rigs:

- 4.4.1** All areas designated as hazardous (classified) locations under the Class and Zone system and areas designated under the Class and Division system shall be properly documented.
- 4.4.2** An approved key plan of the installation showing the different classes of hazardous zones, shown in distinct pattern is maintained.
- 4.4.3** Different classes of zones are distinctly marked at actual places in the installation by suitable notice boards or other effective means; and
- 4.4.4** Every person in the installation is made aware of the dangers associated with and precautions to be taken in different classified hazardous zones.

4.5 Use of electrical equipment in hazardous areas.

Hazardous areas need special types of equipment and installation to ensure safe performance under conditions of proper use and maintenance. Equipment, wiring methods, and installations of equipment in hazardous (classified) locations shall be intrinsically safe, approved for the hazardous (classified) location, or safe for the hazardous (classified) location.

- 4.5.1** In zone '0' hazardous area, no electrical equipment shall be used and where it is not practicable, intrinsically safe apparatus are only to be used which shall be conforming to relevant standards and such installation of apparatus shall conform to relevant standards.
- 4.5.2** The electrical appliance, equipment, machinery or other material that are used in zone '1' or zone '2' hazardous area of the installation shall be of a type and specification conforming to relevant Indian standard or an international standard.
- 4.5.3** Electrical apparatus used in hazardous areas should be manufactured, tested, marked, and installed in accordance with international standards and certified by an independent testing recognized laboratory.
- 4.5.4** Records of certification / test reports conforming to relevant standards from an Indian Govt. Labs or NABL accredited labs or IECEx accredited labs or ATEX notified body which is not a part of manufacturer's facility shall be maintained.
- 4.5.5** Electrical apparatus in hazardous area drilling well and mud processing areas should meet at least Group IIA and temperature class T3.
- 4.5.6** Record of all electrical equipment installed in the Hazardous area shall be maintained.
- 4.5.7** In paint lockers, acetylene stores, and similar spaces where flammable mixtures are liable to collect as well, no electrical equipment should be installed unless the risk/hazard analysis is done.
- 4.5.8** Only explosion proof and intrinsically safe heaters shall be permitted on or near the rig floor, substructure, pits, or cellar.
- 4.5.9** For FLP enclosures, only FLP double compression glands shall be used.
- 4.5.10** Use of non-flameproof equipment / instruments and non-sparking tools is not permitted in hazardous area. However, during S/D or in extreme condition, if non flameproof electrical tools are to be used in Hazardous Area, hot work permit shall be obtained as per Work permit Procedure.
- 4.5.11** Refer **Annexure – 4** of **Part-1** for 'Type of electrical apparatus used in hazardous zones'.

4.6 Maintenance Requirements Electrical Equipment installed in Hazardous (Classified) Locations.

Equipment in hazardous locations shall be maintained such that the following criteria are met:

- 4.6.1** No energized parts are exposed.
- 4.6.2** All bonding jumpers are securely fastened and intact.
- 4.6.3** All fittings, boxes, and enclosures with bolted covers have all bolts installed and bolted tight.
- 4.6.4** All threaded conduits, glands are wrenched tight and enclosure covers are tightened in accordance with the manufacturer's instructions.
- 4.6.5** All fittings, boxes, and enclosures remain fully sealed, with no openings that could weaken their protection.
- 4.6.6** Any alteration to the apparatus / equipment may invalidate certificate related to the equipment / apparatus. If any modifications are carried out the same should be approved by OEM. MOC procedure to be followed as per OISD-STD-178.
- 4.6.7** All parts of electrical equipment installations shall be kept clean and free from accumulation of dust as the same could lead to excessive rise in temperature.
- 4.6.8** The integrity of FLP equipment shall be ensured through inspection at least once in a year.
- 4.6.9** Equipment used in hazardous areas shall be checked and ensured as per OISD-RP-147.
- 4.6.10** The installation shall designate a competent person for inspection and maintenance of flameproof equipment installed in hazardous area location.
- 4.6.11** Regular periodic inspection should be carried out by competent personnel with adequate knowledge of area classification, requirements of Electrical equipment in Hazardous areas and their implications.
- 4.6.12** Switchgear room or control room when located in hazardous areas or its vicinity, should be pressurized or artificially ventilated as per IEC 60079-13 for protection of internal electrical equipment.

5.0 EARTHING& BONDING:

5.1 System Earthing (i.e. an intentional connection of the neutral point of the electrical power supply system to hull or structure)

System earthing shall be considered for all electrical power supply systems to control and keep the system's voltage to earth within predictable limits. It shall also provide for a flow of current that will allow detection of an unwanted connection between the system conductors and earth, which should instigate automatic disconnection of the power system from conductors with such undesired connections to earth.

The selection of one of the following methods of treating the neutral for a specific electrical power system shall be based on technical and operational factors including safety requirements:

- a. directly earthed (TN system)
- b. impedance earthed (IT system)
- c. isolated (IT system)

For Summary of principal features of the neutral earthing methods refer **Annexure-2 (Table-2)**

5.1.1 In mobile offshore rigs, an isolated neutral system is preferred for emergency power system to enhance electrical safety and system reliability. The use of ungrounded systems (floating systems) with

appropriate insulation monitoring devices shall be implemented.

5.1.2 Each insulated, or high resistance earthed primary or secondary distribution system shall have a device or devices to continuously monitor the values of electrical insulation to earth and to give an audible or visual indication in case of abnormally low insulation values. For high & medium voltage system the alarm shall be both audible and visual or provided with automatic disconnection. The standard procedure or Guidance to a system shall be available for the investigation of earth faults.

5.1.3 AC uninterruptible power systems (UPS) shall have an isolated neutral.

5.1.4 Distribution systems with hull return should not be installed, but this does not preclude, under conditions approved with risk analysis and safety consideration the installation of:

- a. impressed current cathodic protective systems;
- b. limited and locally earthed systems e.g. engine starting system;
- c. limited and locally earthed welding systems; where the equipotential of the structure is assured in a satisfactory manner, welding systems with hull return may be installed without this restriction; and
- d. Insulation level monitoring devices provided the circulation current does not exceed 30 mA under the most unfavorable conditions.

5.1.5 In the case of earthed systems, the residual current device (RCD) shall be installed which shuts down the system or the loads if earth fault current is exceeded and prevents a hazard. The settings of protective relays shall be set between 200 to 400 milliampere for individual apparatus of voltage up to 1100 V with suitable time delay protection.

5.2 Equipment Earthing and bonding-

5.2.1 All metallic parts of a unit, that are not normally current-carrying parts, shall be designated as either an exposed conductive part or an extraneous conductive part.

5.2.2 All exposed conductive parts shall be earthed. Earth or an equipotential bonding system may be the steel structure or the hull of a unit.

5.2.3 Extraneous conductive parts shall be connected to an equipotential bonding system.

5.2.4 All metal sheaths and armour of cables should be earthed.

The bonding shall be such as to give a substantially equal potential.

5.2.5 Secondary windings of instrument transformers shall be earthed.

5.2.6 Metal frames or enclosures of equipment mounted in direct metallic contact with the unit structure need no supplementary bonding. The metallic contact shall be clean and free from rust, scale or paint when installed and are firmly bolted together.

5.2.7 Removable gland plates shall be separately bonded to the parent equipment

5.2.8 Every bonding connection to earth shall be of copper or other corrosion resistant material and shall be securely installed and protected against damage.

5.2.9 The nominal cross-sectional area of every copper bonding connection shall be as per **Annexure-2 (Table-1)**

5.2.10 Every connection of an earth conductor or a bonding conductor to the unit structure or hull shall be made in an accessible position and shall be secured by a screw of brass or other corrosion resistant material, which shall be used for this purpose only.

5.2.11 Any electrical or instrumentation equipment attached, but not welded, to the structure steelwork, for example to handrails, ladders, and stairways, shall be bonded to the nearest structural steelwork.

5.2.12 All metal coverings of cables shall be earthed at both ends.

- 5.2.13** The metal covering of cables may be earthed by means of glands intended for that purpose and so designed as to ensure an effective earth connection.
- 5.2.14** Instrument cables without armour shall normally have screens earthed at the control equipment.
- Note - An evaluation must be made regarding the need for earthing in one or both ends of the armour/screen in relation to the required suppression of the frequency band.
- 5.2.15** Electrical continuity shall be maintained at splices between sections of cable ladder, rack or tray by the use of splice plates. Additional bonding is not required, unless cable ladder, rack or tray is insulated from the steel structure or hull to prevent galvanic corrosion.
- 5.2.16** Equipment skids, which are not seam-welded to structural steel, shall be bonded to earth.
- 5.2.17** Electrical continuity shall be maintained between HVAC ducting sections and between ducting and the main structure.
- 5.2.18** Earth continuity of connection shall be tested every six months and records thereof shall be maintained.

6.0 CABLE

6.1 CABLE TRAY PLACING

- 6.1.1** Access for maintenance and an orderly layout shall be ensured. This is also valid when cables are installed below raised floor.
- 6.1.2** Cable ladder systems shall be protected from danger of dropped object due to crane handling or similar.
- 6.1.3** Cable trays must be maintained to fully support the cable installation.

6.2 LAYING AND TERMINATION

- 6.2.1** All cables should be routed on cable ladders or trays.
- 6.2.2** In the Top drive system(TDS), all electrical cables should be laid in service loops /E loops.
- 6.2.3** In the Top drive system, all electrical motors, junction boxes, cabling, Ex connectors etc. and service loops shall be inspected for damage or deterioration and replaced or repaired in accordance with OEM guidelines.
- 6.2.4** Power cables and instrumentation/ control cables shall not be installed on the same cable ladders or trays. Where insufficient space makes this impossible, cables for low voltage, control and instrumentation may be installed on the same tray, but not in the same cable bunch. A separator should be installed on the tray or ladder if different types of cables are installed on the same tray or ladder.
- 6.2.5** Multi-cable transit devices to be used to seal cables entering electrical equipment or enclosures within designated hazardous or classified areas.
- Gas-tight (to maintain barrier for area Classification)
 - Water-tight (to withstand a hydro-static pressure differential)
Primarily encountered within hulls where a relatively high-pressure differential must be withstood
 - Fire-barrier (to withstand fire on the outside from affecting inside)
- 6.2.6** Cables and wiring serving essential, or emergency power, lighting, internal communications, or signals should, so far as practicable, be routed clear of galleys, machinery spaces and their casings and other high fire risk areas. Cables connecting fire pumps to the emergency switchboard should be of a fire-resistant type where they pass through high fire risk areas.
- 6.2.7** Cable runs shall not normally include joints (splices). If, in the case of repair or sectional construction of the unit, a joint is necessary, the joint shall be of such a type that electrical continuity, insulation,

mechanical strength and protection earthing, and fire-resisting or flame-retardant characteristics meets as per design.

6.2.8 All Un-terminated cables must be dead, insulated and clearly identified.

6.2.9 For cable connection to drilling or workover rigs top drive Service Power loops shall be used with Ex plugs and socket for terminating power and control loops.

6.2.10 Explosive Atmosphere Cable Gland for all types of Unarmored & Braid Armour Cables housed in Conduit shall be used for termination cables in motors and lighting or as applicable.

6.2.11 Hazardous area connector (Ex connector) shall be used for low power connections and disconnection safely in hazardous zones for connections.

6.2.12 Ex. Certified cable glands shall be used in hazardous area.

6.2.13 Check the insulation resistance (IR) value of cable insulation for cable annually.

6.3 CABLES AND PLUGS / SOCKETS:

6.3.1 Cables

Selection and installation of cables in hazardous areas should be in accordance with the relevant standard. Electrical cables shall meet the following:

6.3.1.1 All electrical cables and wiring external to equipment shall be at least of a flame-retardant type.

6.3.1.2 Conductor cross sections shall be based on the rating of the overcurrent and short circuit protection used.

6.3.1.3 Only cables associated with type "ia" equipment should be permitted in zone 0 areas.

6.3.1.4 Permanently installed, fixed cable passing through zone 1 hazardous areas shall be fitted with conductive covering, braiding, or sheathed for earth detection.

6.3.1.5 Thermoplastic sheathed cables, thermosetting sheathed cables or elastomeric sheathed cables shall be used for fixed wiring in zone 2 areas.

6.3.1.6 Cables used in drilling and workover rigs should have characteristics such as Flexibility, Drilling Mud Resistance, Chemicals Resistance, Hydrocarbon and Oil Resistance, crush and impact resistance, High pressure gas and steam Resistance, Humidity Resistance, Direct Sun Light resistance and Temperature Shocks Resistance and, they should be Flame retardant and 'Fire resistant as per applicability.

6.3.1.7 Cables regular inspection include visual inspection for damage, electrical continuity testing, and insulation resistance measurement to be done on yearly basis for detection of any damage viz. deterioration due to aging effect, cracking of the insulation, chemicals, water, mud, excessive heat or cold, and mechanical movement effects etc.

6.3.2 Plugs and Sockets

6.3.2.1 No water / moisture shall be permitted in the plug / sockets.

6.3.2.2 Ensure the integrity of plug, sockets, and their connections.

6.3.2.3 If plugs and sockets are present in a hazardous area, they should be suitable for use in the classified zone and have mechanical and/or electrical inter-locking to prevent an ignition source occurring during insertion or removal of the plug.

6.3.2.4 Socket outlet to be installed on external areas shall be of metal which is, or has been rendered, corrosion-resistant, or of flame-retardant insulating material. Where electrolytic corrosion may occur between a metal enclosure and the surface upon which it is mounted, suitable insulating material shall be inserted, and separate means shall be provided to bond the case to earth.

- 6.3.2.5** Each socket outlet or receptacles shall have a plate informing the service voltage and rated current.
- 6.3.2.6** Socket outlets with a rated current above 63amp AC or 16amp DC shall be provided with interlocks so that plug can only be inserted and withdrawn when the switch in “off Position”.
- 6.3.2.7** For rewirable accessories, the contacts shall be indicated by the symbols for phase's, neutral and earth.
- 6.3.2.8** Where socket outlets with earthing are required, the earthing contacts shall make contact before the live contact pins do when inserting a plug.
- 6.3.2.9** Ex. Certified plug & socket shall be used in hazardous area.

7.0 OPERATION, MAINTENANCE AND REPAIR PRACTICES

7.1 GENERAL

- 7.1.1** Recommended practices of OEM of respective equipment and relevant standard shall be followed and records maintained thereof. Inspection shall be carried out and record shall be maintained.
- 7.1.2** The following tools/ Instruments conforming to BIS specifications or equivalent shall be made available, for electrical jobs:
- a) Multi-meter;
 - b) Insulation tester
 - c) Earth tester;
 - d) Tong tester;
 - e) Electrical tool box containing various general tools like Insulated pliers, Insulated screw drivers etc;
 - f) HT Discharge stick;
 - g) Double ladder;(A type insulated)
 - h) Insulated Hand gloves; (As per voltage level) IEC 60903, IS 13774
 - i) Lux meter;
 - j) Cell Tester;
 - k) Tachometer; (optical)
 - l) Thermometer; (infrared/ non-contact)
 - m) FLP Torch;
 - n) Fuse puller
 - o) Non-Contact voltage tester
 - p) Phase sequence Meter(optional)
 - q) Portable Gas Detector with multi / individual function containing LEL/H2S/O2
 - r) LOTO Kit
 - s) Non-sparking tool
 - t) Resistance bank for battery discharge testing

Note: Intrinsically safe tools/instruments shall be used in Zone 1 & 2

- 7.1.3** Authorized person shall supervise the electrical work. No persons shall be allowed to carry out operations, work or testing on any part of the electrical system unless they meet the required competency and are authorized to perform the task. Competency requirements, training, validation, and authorization methodology should be prepared and approved by OIM.
- 7.1.4** Trainings/ awareness program as suggested in OISD-STD-176 should be conducted.
- 7.1.5** Temporary connections, naked joints & wiring shall be avoided. Inspection shall be done by authorized person for unavoidable temporary connections prior to their use to prevent unsafe situations and record

shall be maintained (In the form of permit).

- 7.1.6 On primary side of the transformer, a linked switch with fuses or circuit breaker of adequate capacity shall be provided.
- 7.1.7 Electrical safety shoe shall be worn during electrical O&M activities. Electrical safety shoes shall be as per IS 15298 and OISD-STD-155.
- 7.1.8 Suitable discharge rods shall be used to ensure that the electrical circuit is completely dead before commencing work on it. Non-Contact Voltage tester shall be used prior to discharge any circuit and ensure no voltage before discharging of the circuit.
- 7.1.9 Extra precautions shall be taken when working in damp areas.
- 7.1.10 All electrical faults and defective electrical equipment must be reported.
- 7.1.11 Fire Extinguishers meant for electrical system (CO₂/ Class C fire extinguisher - Dry Powder / clean agent) shall only be used for electrical fires.
- 7.1.12 Firefighting facilities shall be provided as per OISD-STD-189 on "Fire Fighting Equipment for Drilling Rigs, Work Over Rigs and Production Installations".
- 7.1.13 Insulation resistance of motors and other electrical equipment shall be measured regularly with the help of suitable insulation tester once in every six months and records shall be maintained.
- 7.1.14 The effectiveness of switchgear and protective system shall always be kept in working order and shall be checked by calibrating and testing at least once in a year and the result thereof shall be recorded in separate register kept for the purpose and in electronic form:

Wherever numerical relays are being used they shall be checked by testing procedure as per guidelines of original equipment manufacturer or the relevant standards and periodicity of such checking shall be at least once in a year. Refer OISD-STD- 137 Cl.6.3.1 Relay & Meter or its latest version for Protective relay testing/calibration.

- 7.1.15 Functional testing of Earth Leakage Relays (ELR) and RCD (Residual Current Device) shall be ensured once in a year. Record shall be maintained.
- 7.1.16 All interlocks shall be maintained in working condition and inspected as per maintenance schedule.
- 7.1.17 Standard and proper size/rated cables, glands, plugs, sockets, fuses, relays and circuit breakers shall be used.
- 7.1.18 All transformers, batteries, stationary & rotating electric equipment etc. including portable/ hand held electrical apparatus shall be inspected as specified in OISD-STD-137 on "Inspection Of Electrical Equipment" and records maintained thereof.
- 7.1.19 During maintenance, if any equipment is removed from panel etc. which leaves an open cut-out of the equipment, the same shall be promptly sealed. Blanking plates or other means shall be used to ensure that there is no loss of degree of ingress protection. It shall be ensured that all unused cable entries are blanked off.

7.2 Lighting system:

- 7.2.1 Lighting system can be divided into
 - a. General lighting system supplied from the main source of electrical power
 - b. Emergency lighting system supplied from the emergency source of electrical power.
 - c. Escape lighting system supplied from a battery backup source of electrical power.
- 7.2.2 The escape lighting systems shall have a power supply with minimum 30 min backup time supplied by batteries either integrated or have a supply from an emergency generator.

- 7.2.3** Rig lighting and fixtures shall be of appropriate electrical classification for the area in which they are located (refer API 500 and API 505).
- 7.2.4** All Lighting should be provided with lighting transformer so that voltage between the phase does not exceed 250V.
- 7.2.5** Where practicable, lighting fixtures should be installed for easy access by maintenance personnel without the use of portable ladders. If poles are used, the laydown type should be considered.
- 7.2.6** Helideck lighting, NDB, obstruction lights Beacom lights and HAPI lights shall be installed as per DGCA /ICAO/CAP437/ SOLAS guidelines and maintained as per OEM guidelines.
- 7.2.7** Helideck floodlights should be located so as to avoid glare at pilots, and provision should be made for periodically checking their alignment. The floodlight should be such that helideck markings are illuminated and that shadows are kept to a minimum.
- 7.2.8** Omnidirectional red lights should be fitted at locations to provide the helicopter pilot with visual information on objects
- 7.2.9** Structures such as flare booms and towers may be illuminated by floodlights as an alternative to fitting the intermediate red lights, provided that such lights should be arranged such that they will illuminate the whole of the structure and not interfere with the helicopter pilot's night vision.
- 7.2.10** Lighting fittings should be installed so as to prevent temperature rises which could damage the cables and wiring, and to prevent surrounding material from becoming excessively hot.
- 7.2.11** A main electrical lighting system which should provide illumination throughout those parts of the unit normally accessible to and used by personnel should be supplied from the main source of power.
- 7.2.12** The arrangement of the emergency lighting system shall be such that in case of a fire or other casualty in the space or spaces containing the emergency source of power, following shall be adequately illuminated by emergency lighting for a period of 18 h:
- i. Muster station on deck and over sides.
 - ii. All accommodation, stairways and exits, personnel lift cars, and personnel lift trunks.
 - iii. At the survival craft
 - iv. Emergency escape route
 - v. The main generating stations including their control positions.
 - vi. In all control stations and in all machinery control rooms
 - vii. in all spaces from which control of the drilling process is performed
 - viii. at the stowage position or positions for fire-fighters' outfits.
 - ix. on helidecks, to include perimeter and helideck status lights, wind direction indicator illumination, and related obstruction lights, if any.
 - x. The navigation lights, other lights and sound signals, required by the International Regulations for the Prevention of Collisions at Sea, ~~in force~~.
- 7.2.13** The emergency lighting level shall, as a minimum be 30 % of the general lighting level requirement. Emergency lighting level shall not be below the escape lighting level requirement.
- 7.2.14** All light fittings powered through emergency source of electrical power should be marked/labeled as 'E' with red colour code.
- 7.2.15** Inspection of emergency lighting should be carried out monthly using the checklist required by SOLAS regulation III/36 to ensure that they are complete and in good order.
- 7.2.16** General Lighting and escape area lighting illumination level shall be as per **Annexure-3**.

7.3 Electrical Manpower/workforce:

- 7.3.1** Electrical Safety Officer shall be designated for ensuring electrical safety of offshore rig. Electrical Safety Officer shall carry out recommended periodic tests as per the relevant standards and inspect such installations at intervals not exceeding one year and keep a detailed record.
- 7.3.2** The Electrical Safety Officer shall possess a degree in Electrical Engineering with at least 5 years' experience in operation and maintenance of electrical installations or a Diploma in Electrical Engineering with at least 10 years of experience in operation and maintenance of electrical installations.
- 7.3.3** One Electrical Supervisor in each shift of operation/ maintenance shall be appointed. Supervisor must have a valid Electrical Supervisor's Certificate of Competency.
- 7.3.4** The Technicians to assist Engineers or Supervisors shall possess a certificate in appropriate trade with minimum of two years course from an Industrial Training Institute (ITI).
- 7.3.5** The electrical supervisor shall be responsible for the proper performance of the following duties:
- thorough examination of all apparatus, including the testing of earth conductors and metallic coverings for continuity, as often as may be necessary to prevent danger; nullify potential risk and
 - examination and testing of all new apparatus, and of all apparatus, re-erected in the installation before it is put into service in a new position.
- 7.3.6** In the absence of designated electrical supervisor, appoint in writing by OIM a substitute electrical supervisor of equal competency. The electrical supervisor or the substitute electrical supervisor appointed to replace him shall be personally responsible for the maintenance at the offshore drilling and workover rig of a log-book made up of the daily log sheets and the results of all tests carried out shall be recorded in the log-sheets.

8.0 WELL LOGGING OPERATIONS

Precautions during logging operations:

- 8.1** Hold Safety meeting with Offshore Installation Manager, HSE Manager, Logging Team Head and Electrical supervisor.
- 8.2** All the arc/gas welding machines and cathodic protection equipment to be turned off.
- 8.3** Eliminate Sources of Stray Electrical Energy.
- 8.4** Stray Voltage Check. Verify that the voltage between the wellhead, rig, and unit (including generator skid and barge, when applicable) does not exceed 0.25 V
- 8.5** If a helicopter/ Vessel is approaching keep the device in the well till it departs.
- 8.6** Remove or de-energize any rig electrical wiring that might contact the unit, cable, or explosive device during the logging operation. Top drive systems should be electrically isolated in accordance with the manufacturer's procedures. Work Permit procedure as per OISD standard 105 shall be followed.
- 8.7** Radio Transmitters/ Receivers, mobile phones, battery powered equipment and all generators shall be turned off on the installation within 300 m. of work area well before arming the explosive device and shall remain switched off till the device is 70 m down inside the well below the sea floor.
- 8.8** Do not carry out any electrical testing (I.R. tester) discharging etc. during Well Logging operation.
- 8.9** AC power of the logging unit can be restored after running the device inside the well at least ~~400~~ 70 m. below sea floor.
- 8.10** Electrical grounding of wellhead, service unit and rig should be made prior to operating tools using explosives.
- 8.11** The grounding system shall positively electrically bond the electric wireline unit to the casing/ well head blowout preventer and the rig (sheave support system).
- 8.12** For the recommended features and minimum requirements of downhole equipment, pertain to personnel

safety refer API 67.

- 8.13** For the recommended surface equipment to be used at the well site during operations involving explosives devices refer API 67.

For details refer OISD standard 183 on 'Logging operation'.

9.0 CLUSTER WELL LOCATIONS:

Following shall be ensured at cluster well locations:

- 9.1 Prior to switching on power supply, **gas free atmosphere** shall be ensured.
- 9.2 If any Non FLP motor/other electric driven appliances is within 15 m radius from the well to be scrapped by mechanical means, then these motors shall be shut down during the period of scrapping.
- 9.3 No electrical cables shall pass over the producing well. All the cables shall run through the cable tray maintaining a minimum safe distance of 3 meters from the producing well.
- 9.4 Permit shall be taken during scraping operation.
- 9.5 During scrapping, no electrical test shall be carried out using electrical test equipment.

For details refer OISD standard 186 on 'SIMOPS in E&P Industry'.

10.0 WELDING AND CUTTING OPERATIONS:

Common hazards involved in welding/cutting are sparks, molten metal, flying particles, harmful light rays, electric shocks, depletion in O₂ concentration due to generation of toxic gases, etc.

Following safety measures shall be implemented:

- 10.1** No work such as welding or cutting or grinding shall be undertaken in hazardous area unless the area is duly examined and found gas free by a competent person authorized for the purpose and a report of every such examination shall be recorded and maintained. Periodic / continuous testing / detection of hazardous atmosphere shall be done.
- 10.2** Hot work permit shall be obtained before commencement of welding operations as per OISD STD 105. The person who has issued the permit shall explain the content of the hot work permit to the welder concerned and take his signature thereon to that effect before commencement of hot work.
- 10.3** The work piece should be connected directly to Power supply and not indirectly through pipelines/ structures/ equipment etc in line with OISD-GDN-192 on "Safety Practices During Construction" and OISD-GDN-207 on "Contractor Safety".
- 10.4** Three core cable with proper colour code shall be used for connection of single-phase connection welding machine line with (Ref. OISD-STD- 173 on "Fire Protection System for Electrical Installations").
- 10.5** Work shall be carried out under the supervision of competent person.
- 10.6** Welder shall be imparted training about hazards associated with welding and electric shocks.
- 10.7** Ensure use of PPE like welding suits, leather gloves, safety shoes, safety goggles, helmets etc by the welder while working.
- 10.8** Return cable should be connected as close to the welding point as far as possible. (Ref GDN-192 clause no 6.5.1 (V))
- 10.9** Continuous gas monitoring shall be done during welding in hazardous area.

- 10.10** Welding shall not be carried out on tanks and pipelines filled with inflammable material.
- 10.11** Welding machine shall be kept in a safer location and ventilated area and shall be properly earthed.
- 10.12** Before commencing the welding job, ensure that all the cables and earthing connections are as per the standard welding codes / practices. Welding machine/ power supply shall be switched off immediately on completion of the job.
- 10.13** Cables near the welding area shall be covered with fire blanket prior to commencement of the job.
- 10.14** If Welding Transformer is used, it shall be kept in safe area (outside hazardous area), and single length welding cable without any joint, shall be used.
- 10.15** RCD / residual current circuit breaker (RCCB) of suitable rating shall be installed on electrical circuits/ cables feeding power supply to the welding machine.
- 10.16** Appropriate Welding habitat to be made using fire blankets/ curtains, and shields to cover the work area and protect surrounding areas and workers.
- 10.17** Adequate number of suitable types of fire extinguishers and other firefighting arrangements (with firefighters), are made and kept readily available for immediate use, during the HOT work, and at least 30 minutes after the work is completed, and till the work is in hot condition.
- 10.18** During any work of welding, cutting or grinding, the welder shall ensure that all flammable material, oil grease, oil-soaked earth are removed from the area. No match, lighter, or smoking apparatus or any other source capable of igniting flammable gas is present at or around his place of work .
- 10.19** Oil rigs shall have rugged fire and gas detection systems with sensors and alarms, as well as integrated safety and emergency management systems, to ensure complete control and visibility.
- 10.20** Combustible materials shall be removed from a 35-foot radius around the welding site alternatively Fire-resistant covers can be used to protect combustibles
- 10.21** Adequate work area ventilation shall be provided to prevent build-up of flammable items
- 10.22** Hot slag shall be disposed in an approved container.
- 10.23** Workers shall be Trained in emergency procedures and fire extinguishing techniques.
- 10.24** A firewatcher in the work area shall be engaged for at least 30 minutes after work is completed.
- 10.25** Proper ventilation and lighting provided: Where natural ventilation is not available, Ex. fans / smoke excavators are provided. These are also required for speedy dispersal of fumes generated by welding job. Only approved reduced voltage extension Ex. lights (not exceeding 24 volts) are to be allowed for work inside vessels from consideration of personal safety.
- 10.26** Proper means of exit is required in case of emergencies developed on account of the work or otherwise. Availability of an alternate route of escape should be considered.
- 10.27** Area cordoned off and caution boards provided: To prevent any unwarranted entry in the work area and also to caution other personnel taking actions which may endanger people working on the permit job, precautionary tags / boards are to be provided to display like "No Entry" sign on roads or "Caution - Men at Work Inside" etc.
- 10.28** Portable equipment / Hose nozzles properly grounded: As a precaution against static electricity generation, portable equipment / hose nozzles e.g. nozzle of a shot blasting gun, are to be grounded. Use of hydrocarbon pipelines for earthing should be avoided.
- 10.29** No person other than a competent welder duly authorized by the OIM shall carry out welding or cutting work requiring use of flame or electric welding apparatus.

10.30 Refer OISD-GDN-192 & IS 3016 for safe practices to be followed while cutting/ welding.

10.31 Welders on offshore rigs are at potential risk for electrical burns, eye damage from welding arc, health issues from toxic fumes and gases, exposure to dangerous levels of UV and IR radiation, physical injuries, metal fume fever, Electric shock, Poisoning from chromium and nickel in welding fumes, asphyxiation due to working in confined space and hearing loss due to a noisy work environment. Following additional Safety measures to be adopted during welding on Offshore Drilling and Workover Rigs

- i. Welding Equipment shall be suitable for corrosive weather prevalent in Offshore.
- ii. Power source machines must be built to withstand such an environment with rugged bodies made from heavy-gauge stainless steel.
- iii. Diesel generators or diesel-powered welding machine must have extreme-duty protective enclosures to minimize saltwater intrusion. They should be fuel efficient and produce a low noise level to reduce workers' distraction and hearing damage.
- iv. Diesel-powered welding machine should have excellent cold-starting abilities.
- v. Stick welding equipment shall be equipped with a voltage reduction device (VRD) to prevent electrical shocks to welders.
- vi. When welding on facilities with the welding machine located on an adjacent vessel or barge, the facilities should be effectively bonded to the vessel or barge to avoid galvanic corrosion of the facility or the vessel.

11.0 LIGHTNING PROTECTION:

If the metallic structure/steel structure thickness is greater than 4.8 mm thickness, there is no additional lightning protection is required. As per IEC 62305-3.

11.1 Protection against primary structural damage:

- 11.1.1** Lightning protective systems include air terminals, down conductors and earth terminations so installed as to minimize the possibility of voltages being induced in electric cables due to the passage of electric currents.
- 11.1.2** A protective system need not be fitted to a unit of metallic construction, where a low resistance path to earth will be inherently provided by bolted and welded steelwork from the highest point of the unit to earth.
- 11.1.3** The vertical structures (derricks, metallic masts, derrick posts, superstructures, etc.) and metallic structural members may form part or all any protective system.
- 11.1.4** Suitable means shall be provided to enable units, when in dry dock or on a slipway, to have their protective systems or metal hull connected to an efficient earth on shore.
- 11.1.5** An air terminal shall be fitted to each non-metallic mast.
- 11.1.6** Air terminals shall be made of copper or copper alloy. The material shall be resistant to seawater.

11.2 Down conductors

- 11.2.1** Down conductors shall be made of copper or copper alloy, tapes or cables. Cable is preferred as both the insulation and circular shape inhibit surface discharge.
- 11.2.2** The resistance between air terminals and earth terminals shall not exceed 0.02 ohms.
- 11.2.3** A flare boom, drilling rig or crane shall be bonded to the main structure.

NOTE Special consideration must be observed for mobile units during dry docking where the normal connection to earth can be missed. (61892-6)

11.3 Protection against secondary damage

- 11.3.1** Metallic enclosures shall be earthed to the metal structure or hull or to the protective system.
- 11.3.2** Particular attention shall be paid to navigation lights and other equipment at the top of masts and other elevated structures.
- 11.3.3** Lightning earth connections to the protective system shall follow the most direct route.
- 11.3.4** The formation of cable loops, or metallic loops such as pipework, in proximity to down conductors shall be avoided. Cables in close proximity to down conductors shall be installed in metal pipes.
- 11.3.5** On metal units, cabling along decks shall be installed close to the deck to minimize the cross-sectional area of the loop existing between the cable and the deck. When choosing routes along decks, advantage shall be taken of the screening effect of earthed metallic structures near to or above the cable runs, for example handrails, pipes, etc.
- 11.3.6** Means shall be provided for the discharging to earth of any lightning energy that may be induced in, for example, radio and navigational equipment antennas. Consideration shall be given to installing devices such as spark gaps or surge diverters to provide protection from voltage transients.

12.0 PROTECTION AGAINST ELECTRIC SHOCK

- 12.1** All Electrical hand tools shall be double-insulated or grounded in accordance with NFPA 70. Ground fault circuit interruption protection is recommended.
- 12.2** Rig substructure, derrick/mast, and other equipment, enclosures/covers as appropriate grounded to prevent accumulation of static charge
- 12.3** Only authorized personnel, who have undergone approved electrical safety training and those being trained, may perform work on electrical equipment.
- 12.4** Person who is working on an electric supply line or apparatus or both shall be provided with
 - a. personal protective equipment, tools and devices such as rubber gloves and safety footwear suitable for working voltage, safety belts for working at height, nonconductive ladder, earthing devices of appropriate class, helmet, non-contact line tester, hand lines, voltage detector and hand tools as per the relevant standards.
 - b. Arc Flash suits of suitable rating for protection from mechanical and electrical injury due to arc flash.
- 12.5** Electricians must use insulated tools when working inside the restricted approach boundary of exposed energized electrical conductors or circuit parts where tools or handling equipment might make accidental contact.
- 12.6** Exposed metal parts of electrical machines or equipment which are not intended to be live but which are liable under fault conditions to become live should be earthed (grounded).
- 12.7** All electrical tools and portable work equipment must be inspected by an electrical competent person and they shall be clearly identified by a permanent number and the last inspection date.
- 12.8** An approved insulated rescue tool should be available in all switchgear / transformer rooms for the purpose of removing persons from live conductors.
- 12.9** All electrical work should be performed along with one standby person.
- 12.10** All maintenance personnel, emergency response team members and any person acting as a stand-by person for personnel performing electrical work must be familiar with emergency response for electric shock (CPR) and trained in the use of the insulated rescue tool.
- 12.11** All electrical work in designated hazardous areas, where an explosive gas mixture is likely to occur in normal operations, requires a Permit to Work.
- 12.12** When it is necessary to perform maintenance or repair on live electrical systems a Permit to Work is required.



All electrical energy sources must be identified and isolated prior to performing maintenance or repair as per Isolation procedure, permit to work and lockout/tagout procedure of OISD-STD-105.

12.13 Guarding of Energized Conductors and Circuit Parts: Enclosures shall be maintained to guard against accidental contact with energized conductors and circuit parts and other electrical hazards.

12.14 Personal safety and protective equipment such as the following shall be maintained in a safe working condition:

- a. Grounding equipment
- b. Hot sticks
- c. Rubber gloves, sleeves, and leather protectors
- d. Test instruments.
- e. Blanket and similar insulating equipment
- f. Insulating mats and similar insulating equipment
- g. Protective barriers
- h. External circuit breaker rack-out devices
- i. Portable lighting units
- j. Temporary protective grounding equipment
- k. Dielectric footwear
- l. Protective clothing
- m. Non-contact voltage line testers
- n. Insulated and insulating hand tools

12.15 Every installation of voltage exceeding 250 V shall affix permanently in a conspicuous position a Warning signs, danger notices in Hindi or English and the local language, with a sign of skull and bones of a design as per relevant standards

12.16 The use of electricity to electrical installation, shall be controlled by a residual current device to disconnect the supply having rated residual current and duration as per the relevant standards: Provided that in domestic installation viz. Living quarters, Galley equipment, Lighting and Portable tools etc., residual current device having residual operating current not exceeding 30 milliampere shall be used.

12.17 Effects of Electricity on the Human Body.

- a. At 0.5 mA, shock is perceptible.
- b. At 10 mA, a person may not be able to voluntarily let go of an energized electrical conductor or circuit part.
- c. At about 40 mA, the shock, if lasting for 1 second or longer, can be fatal due to ventricular fibrillation.
- d. Further increasing current leads to burns and cardiac arrest.

12.18 Insulating mat shall be provided as per relevant standard in front of switchgear and control gear assemblies and also at the rear, if access from the rear is required.

12.19 The insulating mat or grating shall be oil-resistant and non-slip type.

12.20 Before each use, portable cord- and plug-connected equipment shall be visually inspected for external and internal defects. (NFPA 70 E).

12.21 A single-line diagram, provided for the electrical system, shall be maintained in a legible condition and shall be displayed in PCR or substation /control room.

12.22 Locks, interlocks, and other safety equipment shall be maintained in proper working condition to accomplish the control purpose.

13.0 BATTERY

13.1 Measures shall be taken in stationary battery installations for protection against direct contact and indirect contact or both.

- 13.2** Do not wear conductive materials such as loose jewelry, rings or bracelets which could short circuit the battery causing personal injury, fire or damage to the battery.
- 13.3** The flooded cell batteries should be installed in dedicated rooms.
- 13.4** Batteries are typically installed on open racks. The room is physically separated from other areas, avoiding localized heat sources, with controlled access and doors and partitions determined to meet the required fire resistance rating for the application.
- 13.5** Batteries of the unsealed type shall be in enclosures with outside vents or in well-ventilated rooms and shall be arranged so as to prevent the escape of fumes, gases, or electrolyte spray into other areas.
- 13.6** For safety, Battery room, wherever exist, should be designed with Ventilation systems, forced or natural to prevent buildup of explosive mixtures.
- 13.7** The charging of the battery (particularly those made from lead-acid) releases a dangerous gas: hydrogen. In addition to ventilation, in every battery room the installation of a redundant and ex-rated hydrogen detection system should be provided to ensure personnel and infrastructure safety.
- 13.8** Indoor installations batteries that produce hydrogen or other flammable gases during charging shall meet the exhaust ventilation requirements. The ventilation system shall be designed to limit the maximum concentration of flammable gas to 25 percent of the LFL lower flammable limit (which is 1% hydrogen by volume in air) of the total volume of the room, walk-in unit, enclosure, container, or cabinet during the worst-case event of simultaneous “boost” charging of all the batteries.
- 13.9** Battery room environment must be dry and well ventilated. The ventilation system can be natural or forced, and the selection of the design type will depend on the application.
- 13.10** Only approved insulated tools shall be used to connect or disconnect cells.
- 13.11** Before cells can be disconnected, the battery chargers and the loads must be isolated.
- 13.12** Each battery room or battery enclosure shall be accessible only to authorized personnel.
- 13.13** Battery Activities that Include Handling of Liquid Electrolyte, the following protective equipment shall be available to employees performing any type of service on a battery with liquid electrolyte:
- Goggles and face shield appropriate for the electrical hazard and the chemical hazard
 - Gloves and aprons appropriate for the chemical hazards
 - Portable or stationary eye wash facilities close to the work area that are capable of drenching or flushing of the eyes and body for the duration necessary to the hazard.
- 13.14** Electrical hazard warnings indicating the shock hazard due to the battery voltage and the arc flash hazard due to the prospective short-circuit current, and the thermal hazard Informational warnings signs /Labels shall be displayed at appropriate locations.
- 13.15** Prior to any work on a battery system, a risk assessment shall be performed to identify the chemical, electrical shock, and arc flash hazards.
- 13.16** Batteries for emergency service, including emergency diesel-engine starting, shall be located where they are protected as far as is practicable from damage caused by collision, fire, flooding, spillage, or any other casualty.
- 13.17** Battery bank assembly inside chargers or UPS enclosures should be avoided due to corrosive vapors and possible release of hydrogen.

- 13.18** Ventilated lead-acid batteries and alkaline secondary batteries shall not be placed in the same battery compartment. When different electrolyte type batteries are contained in the same room, precautions and warning labels shall be installed to avoid mixing maintenance tools, electrolyte and topping up water.
- 13.19** A danger notice shall be permanently secured to doors or covers of battery compartments, lockers, and boxes, indicating that any source of ignition in these rooms or in their vicinity is prohibited.
- 13.20** Where movement is possible, in floating installations for example, batteries shall be securely fixed. The trays shall be arranged to give them access to the air on all sides. Any isolating supports shall be non-absorbent to the electrolyte.
- 13.21** Floors shall be of acid-resistant construction unless protected from acid accumulations.
- 13.22** In battery room neutralizing spill kits, a fire extinguisher, and Personal protective equipment should be available.
- 13.23** Accumulator batteries, except for batteries of self-contained battery-operated lights, should not be located in sleeping quarters. Exemption may be granted to this provision based on risk analysis where hermetically sealed batteries are installed.
- 13.24** Make sure the battery charging area is adequately light illuminated and ventilated.
- 13.25** Batteries and battery systems should be maintained and operated in accordance with the following IEEE guides and standards:
- IEEE Std 450 [B32] for VLA batteries
 - IEEE Std 1106 [B35] for Ni-Cd batteries
 - IEEE Std 1188 [B37] for VRLA batteries
- 13.26** Fire detector used in battery rooms shall be intrinsically safe.
- 13.27** The supply / exhaust fans should be of Ex. type and should not produce any static charge.
- 13.28** In the battery room, all electrical fittings, including the light fittings, should be of an explosion-proof type.
- 13.29** Battery stands should be coated to withstand acid or battery shall be placed on acid resistant material.
- 13.30** Hydrogen sensor alarm shall be hooked up to the remote monitoring station.
- 13.31** Battery charging instructions (as per OEM) and MSDS shall be maintained in the battery room.
- 13.32** Battery inspection and testing shall be done as per OISD-STD-137.
- 13.33** Battery room shall have fire resistant doors.

14.0 PRE-COMMISSIONING CHECKS

The Pre-commissioning checks for electrical system of rig prescribed by OEM are to be carried out.

15.0 ELECTRICAL POWER SOURCE

15.1 GENERAL

Electrical installations shall comply with the following:

The Main electrical power source shall be of suitable capacity to be able to handle the load requirement in normal operational condition. The capacity of emergency power source shall not be counted towards maintaining normal operations.

- 15.1.1** Services essential for maintaining safety shall be ensured in the event of failure of the main power source.

15.2 MAIN SOURCE OF ELECTRICAL POWER

- 15.2.1** The main power source of rig shall consist of minimum of two electrical power generating units.
- 15.2.2** The capacity of each generator sets shall be such as to be able to handle the load requirement of the Rig (except the power required for drilling operations) independently.
- 15.2.3** Two distribution transformers shall be installed, and the capacity of each set shall be such as to be able to handle the load requirement of the rig.
- 15.2.4** Every unit shall be provided with a lighting system powered from the main electrical generating source and an emergency lighting system powered by the Emergency generating source. The power distribution system of the two-lighting system shall be totally independent such that any fault or fire or inoperability of one source does not affect the operation of the other source.

- 15.2.5** The main source of electrical power should comply with the following:

The main power source shall be equipped with load shedding arrangement to trip non-essential loads, so as to ensure that units are not overloaded in event only one generating unit is in operation during navigation. The functional test of load shedding logic shall be tested annually once.

15.3 Emergency electrical power source

- 15.3.1** Rig shall be equipped with an emergency source of electrical power, which shall be self-contained.
- 15.3.2** The emergency source of power the changeover arrangement and the emergency switchboard shall be located above the worst damage waterline.
- The Emergency generator shall be supplied by an independent fuel source.
 - The Emergency generator shall in the event of failure of the main power source shall auto start and get automatically connected to the emergency switchboard within a time period of ≤ 45 seconds.
- 15.3.3** When accumulator batteries are used as the source of emergency power, their capability shall be as laid out in MODU code.

15.4 Starting arrangements for emergency generators.

- 15.4.1** Heating arrangements as acceptable to the administration shall be maintained for emergency generators when the ambient temperature is less than 0°C .
- 15.4.2** Emergency generators shall be provided with a minimum of 2 independent starting arrangement. The starting arrangement of emergency generators shall be equipped with stored energy of suitable capacity for providing three consecutive starts, the second starting source shall be capable to provide three starts within a time period of 30 minutes.
- 15.4.3** The Electrical and hydraulic starting system of emergency generators shall be maintained from the emergency switchboard.
- 15.4.4** The starting arrangement, charging arrangement and energy storing devices/ equipment should be located in the emergency generator room and these devices should be solely used for the emergency generator operation.

16.0 EMERGENCY SHUTDOWN SYSTEM

- 16.1** Each unit must be equipped with emergency shutdown system capable of shutting down of the following

systems

- a. Ventilation systems
- b. Main generator prime movers including their ventilation systems.
- c. Emergency generator prime movers.

16.2 The emergency shutdown system shall be located at two different locations and one of them shall be located outside the hazardous area.

16.3 The following system shall be operable after initiation of the emergency shutdown as detailed above.

- a. Emergency lighting at the following locations for a minimum time period of 30 minutes.
 - i. Embarkation station and over their sides.
 - ii. In accommodation alleyways, service areas, stairways, exits, personnel lift cars, and personnel lift trunks.
 - iii. In the machinery spaces and main generating stations including their control positions.
 - iv. In all control stations.
- b. Blow-out preventer control system.
- c. General alarm system.
- d. Public address system
- e. Battery supplied radio communication installations.

17.0 ELECTRICAL ALARMS, CONTROL INDICATORS AND INTERNAL COMMUNICATION

17.1 Offshore Rigs shall be equipped with alarms, controls, indication system and internal communication system in accordance with the recommendations of the IMO in main power rooms, critical motors/MCCs, trip indication alarms.

17.2 There should be a minimum of two independent communication means for Communication between the navigating bridge and the engine control room, one of these communication means should provide visual indication of the orders at both the locations.

17.3 Engineers' alarm: An engineers' alarm operated from the engine control room / maneuvering platform, should be provided , and the alarm should be clearly audible in the accommodation cabin of the engineers.

17.4 A reliable vocal communication system should be provided for communication between the main machinery control station or the propulsion machinery control position, the navigating bridge, the engineer officers' accommodation and, at the central ballast control station on column-stabilized units.

17.5 An alarm system which shall trigger an audible alarm and visual indication in the event of any fault that requires attention, should be provided in the main machinery control station. This system shall also be capable of activating an audiovisual alarm at another normally manned control station.

17.6 The alarm system should be equipped with an automatic changeover system for switching from normal power to emergency/ standby power and have provision for alarm in the event of failure of normal power source.

17.7 Alarm shall not be disabled till acceptance, and the fault indication should remain till rectification of the fault.

17.8 Each Rig should be provided with a general alarm system, and the alarm shall be audible in all accessible parts and open decks. The general alarm system signals should be limited to general emergencies, H₂S, combustible gas, fire alarm, and abandon unit signals.

17.9 A public address system should be provided on each rig. The system should be clearly audible in all areas accessible to personnel during routine operations. Announcements should be possible from at least the following locations Emergency response centre, navigation bridge, engine control room, ballast control station, jacking control station, and near the drillers cabin.

17.10 Internal communication System: Every unit should be equipped with a communication system for exchange and the system should be capable of communication between all areas normally in use during operations.

17.11 In high noise areas Audible signals should be supplemented with visual signals.

17.12 Fire detection and alarm system.

17.12.1 Class approved self-monitoring fire detection and alarm system should be installed in all units.

17.12.2 Accommodation areas should be fitted with smoke detectors.

17.12.3 The fire detection and alarm system should be powered automatically from an emergency power source by a separate feeder if the event of failure of the main power source.

17.13 Flammable gas detection and alarm system

17.13.1 A fixed automatic gas detection and alarm system should be so arranged as to monitor continuously all enclosed areas of the unit in which an accumulation of flammable gas may be expected to occur and capable of indicating at the main control point by audible and visual means the presence and location of an accumulation.

17.13.2 At least two portable gas monitoring devices should be provided, each capable of accurately measuring a concentration of flammable gas.

17.14 Hydrogen sulfide detection and alarm system

17.14.1 Every unit should be equipped with a fixed automatic hydrogen sulfide gas detection and alarm system to continuously monitor at least a minimum of the following areas.

- a. Drilling area
- b. Mud processing area
- c. Well fluid test area.

17.14.2 The system should trigger an audio and visual alarm of the unit and be capable of giving audible and visual alarm at the main control points.

If the alarm at the main control point is not answered within 2 minutes, the main control panel shall automatically activate the toxic gas (hydrogen sulfide) alarm and the helideck status light.

18.0 DOCUMENTATION:

Documentation shall be maintained as detailed out in electrical safety check list as depicted in **Annexure-4**.

19.0 ADDITIONAL NOTE: MODU Code requirements shall be implemented for offshore rigs wherever applicable to ensure compliance with international safety and operational standards.

20.0 REFERENCES:

API RP 500, API RP 505, API 67, API 14FZ MODU code 1989 & 2009 and MODU Jan 2020, SOLAS reg. III/36, IEC 61892 part 2 (Mobile and fixed offshore units – Electrical installations system design), IEC 61892 – 6 (Mobile and fixed offshore units-Electrical installations) , IEC 61892 part 4 for Cable selection and laying, IEC 62305-3, IEC 60903, IEC 60079, NFPA 70E, CEAR 2023.

OISD standards 178, 137, 147, 176, 155, 189, 137, 192, 207, 173, 105.

Recommended Code of practices of ONGC, OIL, RIL, CAIRN.

21.0 ANNEXURES

Annexure-1

(Hazardous area classification) (From MODU code)

1.0 For the purpose of machinery and electrical installations, hazardous areas are classified as below paragraphs 2.0 to 4.0. Hazardous areas not covered (such as, but not limited to, well test equipment areas, helicopter fuel storage areas, acetylene cylinder storage areas, battery rooms, paint lockers, flammable gas or vapors vents and diverter line outlets) in this section should be classified in accordance with Clause 4.3

2.0 Hazardous areas zone 0

The internal spaces of closed tanks and piping for containing active non-degassed drilling mud, oil that has a closed-cup flashpoint below 60°C or flammable gas and vapour, as well as produced oil and gas in which an oil/gas/air mixture is continuously present or present for long periods.

3.0 Hazardous areas zone 1

- 3.1** Enclosed spaces containing any part of the mud circulating system that has an opening into the spaces and is between the well and the final degassing discharge.
- 3.2** Enclosed spaces or semi-enclosed locations that are below the drill floor and contain a possible source of release such as the top of a drilling nipple.
- 3.3** Outdoor locations below the drill floor and within a radius of 1.5 m from a possible source of release such as the top of a drilling nipple.
- 3.4** Enclosed spaces that are on the drill floor and which are not separated by a solid floor from the spaces in paragraph 3.2.
- 3.5** In outdoor or semi-enclosed locations, except as provided for in paragraph 3.2, the area within 1.5 m from the boundaries of any openings to equipment which is part of the mud system as specified in paragraph 3.1, any ventilation outlets of zone 1 spaces, or any access to zone 1 spaces.
- 3.6** Pits, ducts or similar structures in locations which would otherwise be zone 2 but which are so arranged that dispersion of gas may not occur.

4.0 Hazardous areas zone 2

- 4.1** Enclosed spaces which contain open sections of the mud circulating system from the final degassing discharge to the mud pump suction connection at the mud pit.
- 4.2** Outdoor locations within the boundaries of the drilling derrick up to a height of 3 m above the drill floor.
- 4.3** Semi-enclosed locations below and contiguous to the drill floor and to the boundaries of the derrick or to the extent of any enclosure which is liable to trap gases.
- 4.4** In outdoor locations below the drill floor, within a radius of 1.5 m area beyond the zone 1 area as specified in paragraph 3.3.
- 4.5** The areas 1.5 m beyond the zone 1 areas specified in paragraph 3.5 and beyond the semi- enclosed locations specified in paragraph 3.2.
- 4.6** Outdoor areas within 1.5 m of the boundaries of any ventilation outlet from or access to a zone 2 space.

4.7 Semi-enclosed derricks to the extent of their enclosure above the drill floor or to a height of 3 m above the drill floor, whichever is greater.

4.8 Air locks between a zone 1 and a non-hazardous area.

5.0 Openings, access and ventilation conditions affecting the extent of hazardous areas.

Except for operational reasons, access doors or other openings should not be provided between a non-hazardous space and a hazardous area or between a zone 2 space and a zone 1 space. Where such access doors or other openings are provided, any enclosed space not referred to under 3.0 or 4.0 and having a direct access to any zone 1 location or zone 2 location becomes the same zone as the location except that:

5.1 An enclosed space with direct access to any zone 1 location can be considered as zone 2 if:

5.1.1 The access is fitted with a self-closing gastight door opening into the zone 2 space,

5.1.2 Ventilation is such that the air flow with the door open is from the zone 2 space into the zone 1 location, and

5.1.3 Loss of ventilation is alarmed at a manned station.

5.2 An enclosed space with direct access to any zone 2 location is not considered hazardous if:

5.2.1 the access is fitted with a self-closing gastight door that opens into the non-hazardous location,

5.2.2 Ventilation is such that the air flow with the door open is from the non-hazardous space into the zone 2 location, and

5.2.3 Loss of ventilation is alarmed at a manned station;

5.3 An enclosed space with direct access to any zone 1 location is not considered hazardous if:

5.3.1 the access is fitted with two self-closing gastight doors forming an airlock,

5.1.2 the space has ventilation overpressure in relation to the hazardous space, and

5.1.3 loss of ventilation overpressure is alarmed at a manned station.

Where ventilation arrangements of the intended safe space are considered sufficient by the Administration to prevent any ingress of gas from the zone 1 location, the two self-closing doors forming an airlock may be replaced by a single self-closing gastight door which opens into the non-hazardous location and has no hold-back device.

5.4 Piping systems should be designed to preclude direct communication between hazardous areas of different classifications and between hazardous and non-hazardous areas.

5.5 Hold-back devices should not be used on self-closing gastight doors forming hazardous area boundaries.

6.0 Ventilation of hazardous spaces

6.1 Hazardous enclosed spaces should be adequately ventilated. Hazardous enclosed mud processing spaces should be ventilated at a minimum rate of 12 air changes per hour. Where mechanical

ventilation is applied it should be such that the hazardous enclosed spaces are maintained with under pressure in relation to the less hazardous spaces or areas and non-hazardous enclosed spaces are maintained in overpressure in relation to adjacent hazardous locations.

- 6.2** All air inlets for hazardous enclosed spaces should be located in non-hazardous areas.
- 6.3** Each air outlet should be located in an outdoor area which, in the absence of the considered outlet, is of the same or lesser hazard than the ventilated space.
- 6.4** Where the ventilation duct passes through a hazardous area of a higher level, the ventilation duct should have overpressure in relation to this area; where the ventilation duct passes through a hazardous area of a lower level, the ventilation duct should have under pressure in relation to this area.
- 6.5** Ventilation systems for hazardous spaces should be independent from those for non-hazardous spaces.

Note: Refer drilling and work over rig, refer **API RP 505 (Refer clause 11)** recommendation for determining degree and extent of classified locations on Mobile offshore drilling units and **MODU code 1989&2009 Chapter 6** Machinery and electrical installations in hazardous areas for all types of units.

Note: MODUs is to be classified in accordance with API RP500 when the Division system is used and API RP505 Clause 11 (11.1to 11.23,12) when zone system is used.

Annexure-2

Sizes of protective-earthing (PE) conductors and earthing connections

Table -1

Type of earthing connection	Cross-sectional area of associated current carrying conductor	Minimum cross-sectional area of copper earthing connection
1 PE conductor in flexible cable or flexible cord	Any	Same as current-carrying conductor up to and including 16 mm ² , or one-half above 16 mm ² but at least 16 mm ²
2 PE conductor incorporated in fixed cable	Up to and including 16 mm ²	Same as current-carrying conductor up to and including 16 mm ² but at least 1.5 mm ²
a) insulated PE conductor	Over 16 mm ²	50 % of the current-carrying conductor but at least 16 mm ²
b) bare PE conductor in contact with metallic covering	1 mm ² to 2.5 mm ² 4 mm ² to 6 mm ²	1 mm ² 1.5 mm ²
3 Separate fixed earthing conductors	Not exceeding 2.5 mm ² Exceeding 2.5 mm ² but not exceeding 120 mm ² Exceeding 120 mm ²	Same as current-carrying conductor subject to minimum of 1.5 mm ² for stranded earth conductor, or 2.5 mm ² for solid earth conductor One-half of the cross-sectional area of the current-carrying conductor, subject to a minimum of 2.5 mm ² 70 mm ²

Summary of principal features of the neutral earthing methods

Table-2

Means of earthing	Not intentionally earthed "Isolated"	Impedance earthed	Directly earthed
System voltage	All methods are potentially applicable (Note: higher voltage systems are likely to have higher VA earth fault levels, which may make directly earthed connections, or low impedance methods, unattractive)		
Overvoltage	The most significant overvoltage are due to causes not influenced by the method of neutral earthing		
Electric shock risk	All major installations are potentially lethal whatever method of neutral earthing is used		
Use of residual current device for electrical safety	Will normally not function	Use of residual current device with 30 mA operating current should be considered	Acceptable
Use of 3-phase 4-wire supply	Not acceptable	Acceptable	Acceptable

Earth fault current magnitude	Depends on system capacitance but usually very low, e.g. 1 A	Depends on impedance value, typically. 5 A – 400 A	May be up to 50 % greater than symmetrical 3-phase value
Sustained operation with earth fault	Normally possible	May be possible but not advisable, depending on impedance value	Not possible
Minimum earth fault protection required	Alarm or indication	Alarm/indication, earth fault relay, over-current protection, depending on impedance	Over-current protection
Switchgear fault rating	May be rated on normal phase to phase or 3-phase symmetrical fault value		May have to be rated on single-phase-to earth or phase-to-phase-to-earth value
Earth fault location	Faults not self-revealing Shall normally be located manually unless core balance current transformers are fitted	If relays fitted, faults self-revealing. Otherwise, shall be located manually	Faults are self-revealing on over-current
Fire risk	Very low, provided that earth fault current does not exceed 1 A. Prolonged fault may present a hazard	Risk of igniting flammable gases. High impedance faults can lead to burning at fault location.	
Flash hazard (phase-to- earth)	Low----- Increasing----- High-----		
Availability of suitable equipment	Similar generation and distribution equipment is applicable on all systems		Allows use of land-based equipment designed for TN-S systems

Note: For details Refer IEC 61892-2

Illumination Levels for General and Escape Lighting

General lighting illumination levels

AREA	Normal lighting		
	Average illuminance E (lux)	Minimum illuminance E (lux)	Maximum illuminance E (lux)
General outdoor areas	50	20	100
General indoor areas, corridors, accommodation etc.	100	40	200
Stairways	150	60	300
Process areas – occasionally	150	60	300
Process areas – frequently manned	300	150	450
Drill floor	300	150	450
Control rooms – Laboratories	500	250	750
Engine rooms – Pump rooms	200	80	400
Auxiliary engine rooms	200	80	400
Workshops	300	120	600
Switchboard rooms	300	150	450
Offices	500	250	750
Laundry, galley and mess area	300	120	600
OHC	300	120	360
Radio rooms	500	250	750
Battery room, UPS room	200	80	400

Escape lighting illumination levels.

AREA	Escape lighting	
	Average illuminance E (lux)	Minimum illuminance E (lux)
General outdoor escape routes	5	1
General indoor escape routes, corridors – Accommodation, etc.	5	1
Stairway escape routes	5	1
Process areas – occasionally manned	5	1
Process areas – frequently manned	10	2
Drill floor	20	3
Control rooms	250	125
Engine room – Pump room	10	1
Workshops	10	1
Switchboard rooms	10	2
Offices	10	1
Laundry, galley and mess area	5	1
OHC	300	120
Radio room	250	125

Loading stations – muster stations	20	3
Overside (sea level)	15	2
Lifeboat – Life raft stations	20	3
Sea level launching areas	15	2

Note: For details refer applicable MODU Code and IEC 61892-2 for lighting.

DRAFT

ELECTRICAL SAFETY CHECK LIST

1. DISPLAY OF CAUTION BOARDS:

- a) Notice forbidding entry of unauthorized persons in Electrical PCR / Switch Board Room. "ENTRY OF UNAUTHORISED PERSONS STRICTLY PROHIBITED".
- b) Display list of electrical authorized persons in Electrical PCR.

2. DISPLAY AT STRATEGIC LOCATIONS:

- i) Power flow diagram. (SLD)
- ii) Hazardous area classification of the location.
- iii) Electrical Shock Treatment Chart.
- iv) Fire Extinguishers for electrical fire.
- v) Do's & Don'ts.
- vi) Role and responsibility of electrical personnel
- vii) Important/emergency telephone numbers.

3. RECORD KEEPING:

- i) Employee's Bio- Data.
- ii) Shift Roster.
- iii) List of trained personnel
- iv) List of persons with Electrical License.
- v) Accident/incident Records.
- vi) Equipment Logbook & maintenance record.
- vii) Commissioning Record.
- viii) Insulation Resistance Record of cables and equipment.
- ix) Work Permits and LOTO records.
- x) Daily check list of electrical equipment.
- xi) Electrical safety training records. (As per OISD-GDN-176)
- xii) Electrical Design criteria.
- xiii) Hazardous area classification drawing.
- xiv) Testing records of emergency shutdown system and Load shedding logic
- xv) Hazardous area electrical equipment registers and inspection record.
- xvi) Rig Lightning assessment study report
- xvii) Electrical instruments calibration record
- xviii) Insulating monitoring /Earth leakage and protection relays testing records
- xix) Electrical safety audit report
- xx) SMP and SOP for all electrical equipment
- xxi) Illumination Survey records
- xxii) Panels Thermography records
- xxiii) Arc Flash study record

4. USE OF PERSONAL PROTECTIVE EQUIPMENT:
 - i) Insulating Hand Glove specification shall be as per IS 13774
 - ii) Insulating mat specification shall be as per IS 15652(2006).
 - iii) Insulated Shoes.
 - iv) Helmet, safety belt, goggles, masks etc.
 - v) Arc suit
 - vi) Electrical Rescue Rod
 - vii) SCBA (Self Contained Breathing Apparatus)
5. AVAILABILITY OF INSULATED ELECTRICAL TOOLS.
6. AVIATION WARNING LIGHT FITTINGS AT THE TOP OF THE MAST.
7. AVAILABILITY OF THE FOLLOWING **AT SITE AS APPLICABLE:**
 - i) Double ladder.
 - ii) First Aid box and stretcher.
 - iii) FLP torch.
 - iv) Stand by / emergency generator set.
 - v) Emergency lighting inside Control Room.

Annexure-A

Major Changes in Revised Edition

1. Standard scope is extended to cover Offshore electrical safety for drilling and work over rig.
2. Definitions modified as per the latest CEA Reg. 2023 and added new definitions for onshore & offshore part.
3. New Clause added for use of Electrical equipment in Hazardous area.
4. Additional sub clauses under clause for Cable & Cable laying; Operation, maintenance and repair practices; Welding operation; Logging and Cluster well locations.
5. Additional clauses under Rig Transportation for clearance in air for lowest conductor of overhead lines.
6. New clause Electrical Manpower requirement
7. New clause of Lightning protection
8. New clause of protection against Shock
9. New clause of Power Source
10. New Clause of Emergency Shutdown
11. Additional Annexures:
 - a. Hazardous area classification
 - b. Periodic inspection checklist of FLP equipment
 - c. Illumination Levels on Drilling and WOR.
 - d. Type of protection for electrical equipment in classified areas.

Annexure-B

FUNCTIONAL COMMITTEE

(First Edition - September 2007)

NAME	ORGANISATION
LEADER	
Dr U.S. Hazra	Oil & Natural Gas Corporation Ltd
MEMBERS	
Shri S.V. Havirayaji	Oil & Natural Gas Corporation Ltd
Shri C.M. Sharma	Oil Industry Safety Directorate
Shri A.K. Acharya	Oil India Ltd.
Shri Sathees Kumar	Oil India Ltd.
Shri H. Kumar	Oil & Natural Gas Corporation Ltd
Shri S. De	Oil & Natural Gas Corporation Ltd Shri
Pradeep Kumar	Oil & Natural Gas Corporation Ltd
MEMBER COORDINATOR	
Shri D.K. Varshney	Oil Industry Safety Directorate

In addition to the above, various other experts from the industry contributed in the preparation, review and finalisation of this document.